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Development of Control Strategies for Earthquake Reproduction on Shaking Platform

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Aos meus pais, que me deram a possibilidade de estudar,
Eternamente grato

Resumo

Na área científica da engenharia sísmica, apesar dos desenvolvimentos em simulação computacional, ainda se enfrentam limitações na modelação precisa de comportamentos estruturais complexos, não lineares, ou outras dinâmicas pouco estudadas. As simulações numéricas, mesmo com métodos avançados, requerem modelos experimentais para garantir uma boa precisão. Além disso, as abordagens de aprendizagem automática também dependem de dados experimentais para treinar e atualizar modelos. Os ensaios experimentais permitem por isso a validação e calibração desses modelos. Em resumo, os ensaios experimentais sísmicos continuam a ser indispensáveis para validar modelos, compreender comportamentos complexos e garantir a segurança estrutural.

Nesta tese, um modelo computacional da plataforma sísmica uniaxial do LNEC é utilizado para comparar o desempenho, em termos de espectro de resposta, de três estratégias de controlo distintas: o método iterativo atualmente utilizado, chamado de OLI, que consiste num controlo pseudo-adaptativo offline em anel aberto e com o ensaio a decorrer em anel fechado com um controlador PI (proporcional-integral), um controlador sintonizado PID-F (proporcional-integral-derivativo-filtrado), e um controlador ótimo de formulação linear-quadrática-integral (LQI).

Adicionalmente, para a finalidade de se sintonizar os controladores, é proposto um método de identificação próprio para sistemas que operam apenas em anel fechado, como é o caso das mesas sísmicas do LNEC. Este método é aplicado na identificação de um agitador eletrodinâmico, sendo o modelo identificado usado para comparar, novamente, as três estratégias de controlo. Adicionalmente, é implementado um processo de otimização dos parâmetros da função de custo para o LQI.

Os resultados indicam que ambos os controladores PID-F e o LQI oferecem desempenho superior quando comparados com o método OLI, sendo o controlador LQI aquele com melhor desempenho em todas as simulações. No entanto, o desempenho de cada método depende da identificação do sistema, e deve ser avaliada a sensibilidade de cada método a este fator.

Palavras-chave: Engenharia Sísmica, Controlo de Sistemas, Identificação de Sistemas, Identificação em Anel Fechado

Abstract

Despite computational models used in earthquake simulations becoming highly sophisticated and reliable, they still face limitations when complex, nonlinear, or poorly understood structural behaviours are considered. Moreover, numerical simulations, even with advanced methods, require experimental benchmarks to ensure accuracy, and machine learning approaches also rely on experimental data for training and updating models. Experimental testing provides critical validation and calibration for these models. In summary, earthquake experimental testing remains indispensable for model validation, understanding structural complex behaviours, and ensuring structural safety.

This thesis analyses control strategies for reproducing earthquake ground motion on the uniaxial shaking table located at Laboratório Nacional de Engenharia Civil, through a computational model of the system. The objective of the controller is to ensure the shaking table reproduces the target earthquake ground motion as accurately as possible. The overall performance is evaluated in terms of response spectra for three distinct control strategies: the currently used OLI method consisting on pseudo-open loop offline adaptive control with an iterative process of generating drive signals and running the tests in closed loop with a PI controller, a tuned PID-F controller (proportional, integral, and filtered derivative), and a LQI controller.

Additionally, in order to tune the controllers, an identification method more appropriate for systems operating only under closed-loop, such as LNEC's shaking tables, is proposed. This method is then applied on the identification of an electrodynamic shaker, whose identified model is then used to, once again, compared the performance of the three control strategies.

The results indicate that both the PID-F and LQI controller offer superior performance compared to the OLI method, whereas the optimal controller performed the best. The performance of the controllers depends on system identification, and the sensitivity of each method to the quality of the identification requires further investigation.

Keywords: : Earthquake Engineering, Control Systems, System Identification, Closed-loop Identification

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Nomenclature

Glossary

1 Introduction

1.1 Seismic Platform Tests at LNEC

The current seismic platform testing process at Laboratório Nacional de Engenharia Civil (LNEC) consists of iteratively synthesizing a drive signal, distinct from the reference signal, so that the platform replicates the response spectra of the reference signal, i.e. the earthquake.

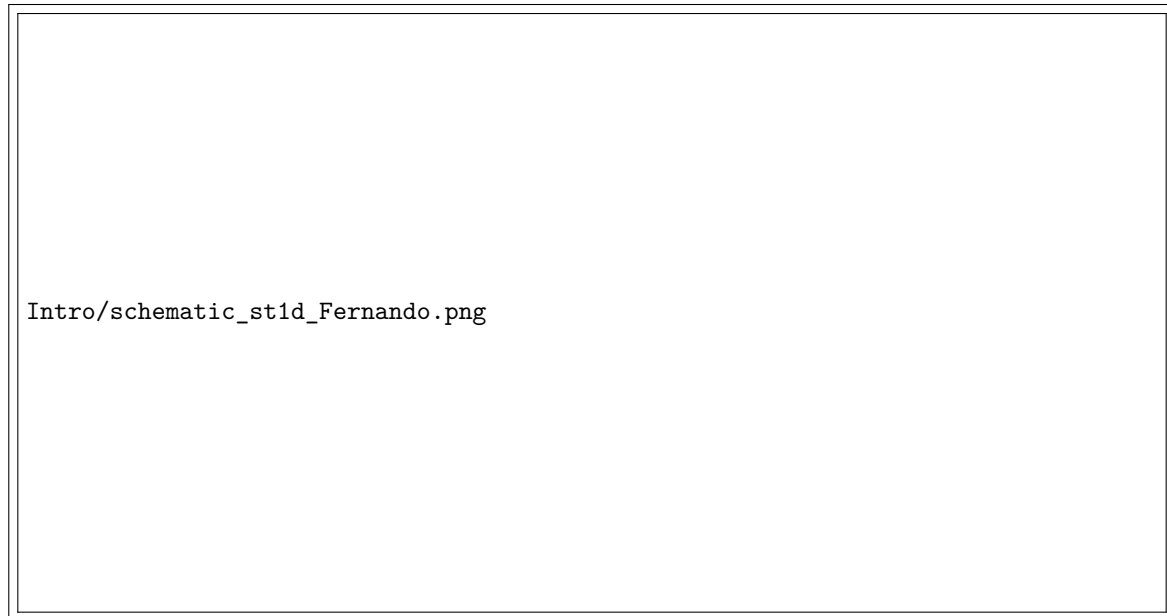


Figure 1: Complete schematic representation of uniaxial shaking table (1D-ST) with a coupled 2DOF system [Oliveira, 2015]

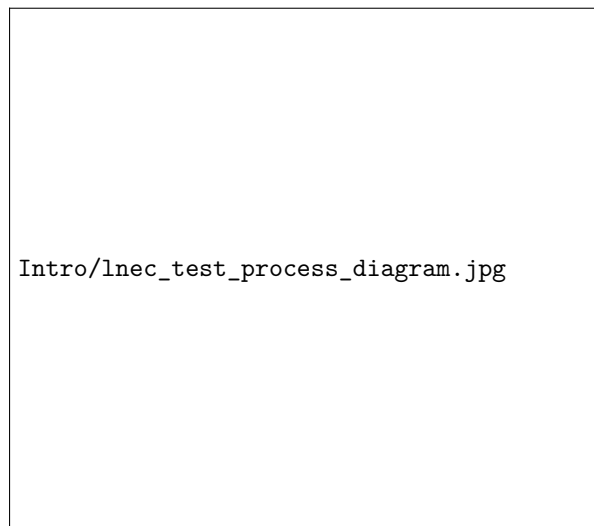


Figure 2: Work flow diagram of the seismic testing process at LNEC. Controller settings (proportional and integral gains) are set beforehand. Additional identifications (before target motion intensity is increased) may be done, particularly when it's verified that the adapted drivers are overcompensating certain frequency bands, and this behaviour is suspected to result from poor Identification.

During the seismic test, the structure is subjected to a series of tests with different earthquake amplitudes. Initially, the test is carried out with the earthquake at a reduced amplitude, assessing the subsequent damage. The procedure is then repeated with the same earthquake, but at a higher amplitude, again assessing the damage, and so on until a defined amplitude is reached, which usually corresponds to the original amplitude of the earthquake.

Before each test, an identification of the system's transfer function is carried out, using a limited-band pink noise signal (pink noise has a slope of -10dB/decade). This function is then used in the command signal generation algorithm. Throughout this process, the amplitude of the reference signal increases progressively, and it is expected that the structure will enter a non-linear regime, either due to the behaviour of some of its components or the damage accumulated during the various tests.

At the end of each test, the system's performance is evaluated by comparing the response spectrum of the reference signal with that of the signal measured on the platform.

However, it is not always possible to achieve satisfactory performance with the command signal synthesized initially, and it may be necessary to generate more than one command signal and carry out additional tests at the same amplitude. This procedure, although necessary in some situations, should be avoided as it induces additional damage to the structure.

In addition, the system faces physical limitations inherent to the equipment, namely with regard to the finite field of displacement, velocity and acceleration, which must be duly considered during the tests.

1.1.1 Response Spectrum

The response spectrum is a fundamental concept in earthquake engineering and structural dynamics. It represents the maximum response (such as displacement, velocity, or acceleration) of a structure modelled as a single-degree-of-freedom (SDOF) system subjected to a ground motion.

It's relevance for this work lies in the fact that the performance of the seismic table, will be assessed mostly based on its ability to reproduce seismic ground motion response spectra, which will be evaluated for frequencies between 0.1 and 30 Hz.

Figure 3 helps us easily comprehend how a response spectrum is determined. The peak amplitudes ③, from the response of 1-DoF mass-spring-damper systems with 5% damping and different natural frequencies ②, to the seismic signal ①, are plotted for each natural frequency ④.

Some fundamental concepts on response spectrum are:

- Ground Motion Dependency
 - It is derived from an earthquake acceleration time history.
 - Different earthquakes produce different response spectra.
- Structures Dynamic Properties
 - The response depends on the structure's natural frequency and damping ratio.
 - Structures with different natural periods respond differently to the same ground motion.
- Spectral Parameters The response spectrum can show maximum values of:
 - Spectral Acceleration: Used for force-based design.
 - Spectral Velocity: Important for energy-based considerations.
 - Spectral Displacement: Relevant for displacement-based design.

Some of its applications in Civil Engineering are:

- Seismic Design of Buildings and Bridges: Response spectra help in defining design forces.
- Code Based Seismic Analysis: Building codes (e.g., *ASCE 7*, *Eurocode 8*, *IS 1893*) provide standard response spectra.
- Performance Based Seismic Design (PBSD): Uses response spectra to predict structural behaviour.

response_spectrum_schematic.png

Figure 3: Schematized process of determining a response spectrum

1.2 Objectives

The objective of this work is to implement a controller, or strategy, to ensure that the seismic platform reproduces the test seismic signal (reference/earthquake) as accurately as possible, taking into account the following aspects:

- The amplitude of the earthquake increases progressively with each test during the experimental campaign;
- Before each test, an identification of the system is carried out at a reduced amplitude (to avoid damaging the model). One objective will be to assess the feasibility of using the earthquake response to identify, or complement the identified model, and cover higher amplitudes.
- The command signal, generated offline, is imposed on the seismic platform without prior evaluation, by simulation, as to its effectiveness in creating the desired output at the seismic platform;
- The seismic platform system with 1 degree of freedom (ST1D) is capable of testing models with a mass of up to 5 tons, and operates in a band of interest from 0 to 25 Hertz. It also has the following limitations (which should be confirmed in laboratory tests):
 - Maximum displacement, in absolute value, of 100 mm
 - Maximum speed, in absolute value, of 0.4 m/s
 - Maximum acceleration, in absolute value, of 25 m/s²
- The model to be tested may initially exhibit linear behaviour, but as it degrades, it inevitably becomes non-linear. In general, the structures tested have two vibration modes in the seismic platform's band of interest, whose initial characteristics (without damage) are:
 - ◊ $1.5 \leq f_1(\text{Hz}) \leq 4$
 - ◊ $2 \leq \xi_1(\%) \leq 5$
 - ◊ $6 \leq f_2(\text{Hz}) \leq 10$
 - ◊ $5 \leq \xi_2(\%) \leq 15$In damaged structures, damping can reach:
 - * $\xi_1 = 10\%$
 - * $\xi_2 = 25\%$
- Performance will be evaluated by comparing the response spectrum of the measured signal with that of the reference signal.

1.3 Outline of Dissertation

Taking into account the objectives

2 Literature Review on Experimental Testing in Seismic Engineering

2.1 The Purpose of Experimental Testing

While computational simulations have become highly sophisticated and efficient, they still face limitations in accurately modelling complex, nonlinear, or poorly understood behaviours, especially in components with significant uncertainties or novel materials. Studies show that numerical simulations, even with advanced methods, require experimental benchmarks to ensure accuracy, especially for complex interactions. Machine learning approaches also rely on experimental data for training and updating models. Experimental testing provides critical validation and calibration for these models.

In summary, seismic experimental testing remains indispensable for validating models, understanding complex behaviours, and ensuring structural safety. The trend is toward integrated approaches using both experimental and computational tools to achieve the most reliable seismic performance assessments. [Williams and Blakeborough, 2001, Zhang and Wu, 2023, Liu et al., 2020, Bahraq et al., 2021, Malomo and DeJong, 2022, Qiu and Elgamal, 2024, Yavas et al., 2023, Mokhtari and Imanpour, 2023]

2.2 An Overview of Experimental Methods

According to Williams and Blakeborough [2001], the most widely used experimental methods in the field of seismic engineering are:

- **Shaking Table Tests:** Shaking tables simulate real earthquake ground motions on structural or soil models, allowing direct observation of dynamic response and failure mechanisms. They are considered the gold standard for dynamic testing of buildings, bridges, tunnels, and other infrastructure. They are used for both full-scale and scaled models, including innovative in-situ testing with mobile shaking tables [Williams and Blakeborough, 2001, Bousias, 2017, Abovyan et al., 2024, Tsinidis et al., 2020, de Felice et al., 2016, El-Emam, 2018].
- **Pseudo-Dynamic Testing and Hybrid Simulation:** In pseudo-dynamic testing, loads are applied over a greatly expanded timescale with the dynamic behaviour of the structure accounted for computationally. In hybrid simulation, just like pseudo-dynamic testing, a combination of physical testing and numerical modelling is used, but the experimental part of the process is performed at the correct rate so that the test specimen can respond dynamically. These methods are widely used for complex or large structures where full dynamic testing is impractical [Williams and Blakeborough, 2001, Bousias, 2017].
- **Static and Cyclic Loading Tests:** These tests apply static or cyclic loads to materials, structural elements, or systems to study their properties under earthquake-like conditions [Williams and Blakeborough, 2001, Bousias, 2017, Fiorino et al., 2016].
- **Centrifuge Modelling:** Centrifuge tests use scaled soil-structure models spun at high speeds to replicate gravitational stresses, enabling realistic simulation of soil-structure interaction during earthquakes, especially for underground structures such as tunnels. This method is mostly focused on testing soil properties, not structures [Williams and Blakeborough, 2001, Tsinidis et al., 2020].
- **Surface Based Seismic Methods:** Used for site characterization, through passive-source (microtremor) and active-source techniques, to determine ground shear-wave velocity profiles [Luke et al., 2008].
- **Laboratory Fault Slip Experiments:** Used to study fault mechanics and induced seismicity, employing shear-flow, triaxial, and injection-driven tests [Ji et al., 2022, Dong and Luo, 2022].
- **Vibration Machines:** Employed for simulating seismic impacts on buildings and models, offering a cost-effective alternative for certain applications [Abovyan et al., 2024].
- **Explosive Charges:** Arrays of explosive charges used to simulate earthquake ground motions at large scale (very uncommon method) [Williams and Blakeborough, 2001].

2.3 Seismic Table Control Strategies

Shaking-table tests consist of mounting a physical model on a stiff platform that is driven so as to reproduce the desired ground-motion time history (reference signal). This experimental technique was introduced in the 1940s and became widespread during the 1960s. Early systems relied on mechanical profiled cams to generate the excitation, but servo-hydraulic systems soon became the standard solution ([1],[2],[3]). At the Laboratório Nacional de Engenharia Civil (LNEC) a Goodmans model VG-108 electrodynamic shaker attached to a movable table was installed in the 60s to perform seismic tests [4]. A uniaxial shaking table (ST1D) was developed and installed, consisting of a $3\text{ m} \times 2\text{ m}$ horizontal steel platform and a vertical platform body driven by a hydraulic actuator supplied by a dedicated oil-hydraulic circuit ([5], [6]). During this period, adaptive control strategies were developed, implemented, and validated both on a small scale shaking table and on the ST1D, for tracking reference earthquake spectrum (frequency domain), iteratively ([7],[8]). Increasingly demanding requirements for reproducing seismic excitation along all three directions motivated the development of the triaxial shaking table (ST3D), which has been installed and operational at LNEC since the mid-1990s [4]. The reproduction of earthquakes on a shaking table depends strongly on: i) the hardware (mechanical and electrical system), which usually has limitations and therefore can distort the reference signal and; ii) the control system, whose performance is highly sensitive to the adopted control strategy and tuning. The hydraulic and control system supplier, MTS Systems Corporation (MTS), provided several control strategies: i) Three-Variable Control (TVC), also known in control theory literature as state variable control [9], which combines the displacement, velocity and acceleration feedback [1]; ii) Adaptive Harmonic Cancellation (AHC), suitable for sinusoidal signals (fixed frequency or sweep); iii) Adaptive Inverse Control (AIC) [10], suitable for non-sinusoidal signals on linear systems; iv) Amplitude/Phase Control (APC), suitable for sinusoidal signals (fixed frequency and sweep) and predominantly linear systems, and often used in conjunction with AHC to reduce distortions in nonlinear systems; v) Online Iteration (OLI), suitable for nonlinear systems and non-sinusoidal waveforms. This last strategy is the most appropriate for earthquake engineering applications, as structures may exhibit nonlinear behaviour or sustain damage during testing. OLI is currently the control method implemented at LNEC's ST3D. Shaking-table control strategies typically begin with the offline generation of a drive signal using the inverse model of the system and the desired reference motion. This drive signal is then applied to the system, which usually incorporates a linear controller, most commonly a Proportional-Integral-Derivative (PID) controller, using actuator feedback signals (displacement, velocity, or acceleration) ([1], [11]). This is the basis of the OLI control strategy. Indeed, PID controllers remain widely used in most test systems, including shaking-table and hybrid testing platforms [12]. Alternative control approaches have been proposed in the literature, including: i) Acceleration Trajectory Tracking Control (ATTC) [13]; ii) displacement feedforward-feedback controller using an optimal controller [14]; iii) Model-based feedforward-feedback controller for acceleration tracking [15]; iii) minimal control synthesis (MCS) strategy, which is feedback-feedforward adaptive control strategy including the inverse model in line with the reference signal [16].

2.3.1 Innovations in Recent Literature

Advanced control methods, such as active disturbance rejection control (ADRC), adaptive model-based state estimators, and nonlinear signal-based control, have significantly improved the accuracy and stability of shaking table operations, especially for reproducing complex seismic waveforms and handling nonlinear structural responses. Offline iterative control and frequency-splitting techniques enable double-layer shaking tables to achieve both high payload and wide frequency range, addressing the needs of testing rigid structures like dams and nuclear facilities. Proportional-derivative (PD) control and affordable hardware have made seismic table testing more accessible in resource-limited settings.

Adaptive control has been shown to significantly improve the accuracy of shaking table tests for dynamic structural evaluation:

- **Enhanced Tracking and Stability:** Adaptive control methods, such as adaptive-model-based state estimators, model reference adaptive control, and adaptive sliding mode control, have been shown to improve both the stability and tracking accuracy of shaking tables, especially under varying test conditions and nonlinear system behaviours
- **Compensation for Uncertainties:** Adaptive controllers can dynamically adjust to compensate for

amplitude and phase deviations, system nonlinearities, and parameter uncertainties, leading to more precise seismic waveform reproduction and better handling of control-structure interaction effects

- **Superior Performance over Conventional Methods:** Comparative studies demonstrate that adaptive control strategies outperform traditional fixed-parameter controllers (such as PID or three-variable controllers) in both time and frequency domains, providing better acceleration tracking and robustness to specimen uncertainties
- **Practical Validation:** Experimental and simulation results across multiple studies confirm that adaptive control approaches—such as online adaptive compensation, adaptive inverse control, and functional link adaptive controllers—consistently yield higher fidelity in replicating target motions and structural responses during shaking table tests

3 System Identification & Control

This chapter presents the theoretical foundations underlying the system identification and control strategies employed in this work. It is organized into three main parts. The first part, Section 3.1, establishes the signal and system analysis framework, covering the definitions, notation, and properties of signals processed by discrete linear time-invariant (LTI) systems. The second part, Section 3.2, addresses control system design, progressing from open-loop to closed-loop architectures, and introducing the specific controllers considered in this work: a PIDF controller, and full state feedback control with optimal control formulation, including the concepts of controllability, observability, and state estimation. The third part, Section 3.3, tackles the problem of system identification in closed-loop settings, discussing the limitations of classical spectral analysis methods and presenting the Two-Stage Method as a suitable approach for closed-loop identification.

3.1 Signals & System Analysis

3.1.1 General Definitions & Notation

For a signal discretized with sampling time T_s :

- The delay operator z^{-1} corresponds to a delay of one sampling time
- The forward operator z corresponds to a forward time shift of one sampling time
- A delay/forward shift of k sampling times T_s corresponds to z^{-k} or z^k , respectively
- $G(z)$, in upper case, designates a transfer function, while the lower case $g(k)$ is the corresponding pulse response at time instant $t = kT_s$, and both are related by:

$$G(z) = \sum_{k=-\infty}^{\infty} g(k) z^{-k} \quad (3.1.1)$$

- The complex-valued function $G(e^{i\omega})$ is the evaluation of the transfer function $G(z)$ on the unit circle $z = e^{i\omega}$ in the complex plain, and is referred to as the frequency response of a discrete time system. This differs from continuous time systems where a frequency response $G(i\omega)$ of a given continuous transfer function $G(s)$ is given by its evaluation on the imaginary axis $s = i\omega$.
- Various equations will have the notation eased by omitting the function variables to improve readability
- The circumflex denotes an estimated model or property from some data (ex. $\hat{G}(z)$ or $\hat{\Phi}_u(\omega)$)

3.1.2 Properties of signals processed by discrete LTI systems

For time signals $x(t)$ and $y(t)$ the correlation is given by:

$$R_x(\tau) = E[x(t)x(t-\tau)] \quad (3.1.2)$$

$$R_{yx}(\tau) = E[y(t)x(t-\tau)] \quad (3.1.3)$$

And the power spectral density:

$$\Phi_x(\omega) = \sum_{\tau=-\infty}^{\infty} R_x(\tau) e^{-i\omega\tau} \quad (3.1.4)$$

$$\Phi_{yx}(\omega) = \sum_{\tau=-\infty}^{\infty} R_{yx}(\tau) e^{-i\omega\tau} \quad (3.1.5)$$

For an LTI system as in figure 4, where $y(t) = G(z) \cdot x(t)$, with x and y quasi-stationary signals, we have the following properties:

$$\begin{aligned} R_{yx}(\tau) &= G(z) R_x(\tau) \\ &= g(\tau) \star R_x(\tau) \end{aligned} \quad (3.1.6)$$

$$\begin{aligned} R_{xy}(\tau) &= G^*(z) R_x(\tau) \\ &= g(-\tau) \star R_x(\tau) \end{aligned} \quad (3.1.7)$$

$$R_{yy}(\tau) = G^*(z) G(z) R_x(\tau) \quad (3.1.8)$$

$$= g(-\tau) \star g(\tau) \star R_x(\tau) \quad (3.1.9)$$

$$\Phi_{yx}(\omega) = G(e^{i\omega}) \Phi_x(\omega) \quad (3.1.10)$$

$$\Phi_{xy}(\omega) = G^*(e^{i\omega}) \Phi_x(\omega) \quad (3.1.11)$$

$$\begin{aligned} \Phi_y(\omega) &= G(e^{i\omega}) G^*(e^{i\omega}) \Phi_x(\omega) \\ &= |G(e^{i\omega})|^2 \Phi_x(\omega) \end{aligned} \quad (3.1.12)$$

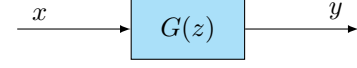


Figure 4: System

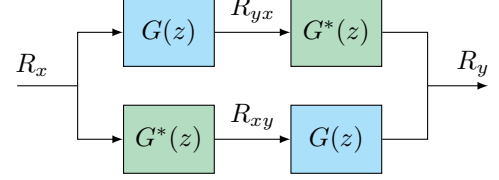


Figure 5: Illustration of how linear systems transform correlation signals

In the case of two systems which generate an output from a common input signal, as in figure 6, then y_2 can be given as function of y_1 , and we have the following relations:

$$y_2(t) = G_2(z) x(t) = G_2(z) G_1^{-1}(z) y_1(t) \quad (3.1.13)$$

$$\Phi_{y_2 y_1}(\omega) = G_2 G_1^{-1} \Phi_{y_1}(\omega) = G_2(e^{i\omega}) G_1^*(e^{i\omega}) \Phi_x(\omega) \quad (3.1.14)$$

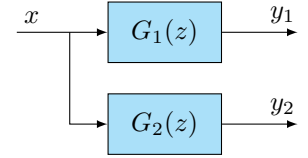


Figure 6: Two systems with a common input signal



Figure 7: Visualization of convolution, cross-correlation, and autocorrelation. The output of the computation is represented by the shaded area, and is shown for 5 different lag values. (Adapted from original of Cmglee [2012])

3.2 Control Systems

There are two generic configurations for control systems: open-loop control (figure 8) and closed-loop control (figure 9).

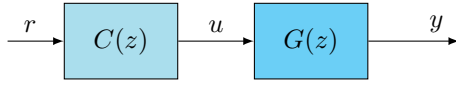


Figure 8: Control in open-loop

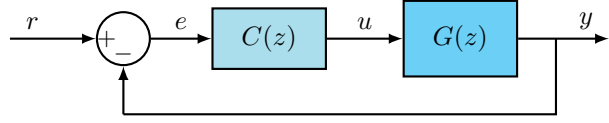


Figure 9: Control in closed-loop

Every control system is a variation and/or a combination of these two configurations. Both add, to the system we want to control, another system called controller which outputs a control action $u(t)$, which is the controlled system's input intended to make the controlled system's output $y(t)$ follow some specified reference $r(t)$ (which is just the desired output). In a perfectly controlled system the output will match the reference, $y(t) = r(t)$.

This section aims to give the reader an overview of the essential control configurations, their advantages and limitations, as well as controller structures and design methods. Open-loop and closed-loop architectures are addressed, with the latter including classical output feedback control with a PIDF, and then full state feedback with optimal control and the underlying concepts of controllability, observability, and state estimation. This segment is specially influenced by the work in Valério [2025] and Botto [2021].

3.2.1 Open-Loop Control

In open-loop control the controller receives the reference that the system should follow and computes a control action. It is not checked whether the system's output then follows the reference, and the control action is not modified to account for any deviations that occur.

For an open-loop control system we have the following equation:

$$y(s) = G(s) u(s) = G(s) C(s) r(s) \quad (3.2.1)$$

The main advantages of open-loop control are:

- It is conceptually simpler than closed-loop control
- It does not require real-time measurement of the output and consequently no sensor for the output variable is required to implement it
- If the reference is known in advance, the control action can be computed a priori

Open-loop control works well if two conditions are simultaneously met: A perfect model of the plant exists and there are no unknown disturbances of either the control action or the output. In practice, open-loop control does not work well due to imperfect modelling of the plants, the existence of process and measurement disturbances, and the fact that it cannot control unstable plants.

3.2.1.1 Design of Open-Loop Controllers

Since we want the output to match the reference:

$$r(s) = y(s) = G(s) C(s) r(s) \implies C(s) = \frac{1}{G(s)} \quad (3.2.2)$$

Thus, the controller should be an inverse model of the system to control. This is problematic if the model of the system is proper, because then the controller isn't, resulting that additional poles must be added to make the controller proper. These additional poles should be of very high frequency as to preserve the dynamics of the idealized controller (which is just the inverse of the plant) such that it cancels out the dynamics of the plant, and the output matches the reference.

Figures/Tecnico_ID_Principal_RBG-COR.png

Figure 10: Open-loop control example: Reference & Output

Figures/Tecnico_ID_Principal_RBG-COR.png

Figure 11: Open-loop control example: Control action

3.2.2 Closed-Loop Control

Closed-loop control uses feedback, meaning the output is being measured and compared with the reference. The difference between the reference and the measured output is the tracking error $e(t)$. In closed-loop control, the tracking error is the input to the controller:

$$y(s) = G(s) u(s) = G(s) C(s) e(s) = G(s) C(s) [r(s) - y(s)] \quad (3.2.3)$$

The simplest form of closed-loop control is with a proportional controller, $C(s) = K$, meaning the control action is proportional to the tracking error:

$$u(s) = C(s) e(s) = K e(s) \quad , K \in \mathbb{R} \quad (3.2.4)$$

Thanks to output feedback, closed-loop control can mitigate the effects of disturbances and imperfect models, since the control action is computed from the tracking error, and thus compensates for any deviations from the reference.

Closed-loop control can also be used to stabilise unstable plants, and the opposite is also true: Badly designed closed-loop control can destabilise stable plants.

Figures/Tecnico_ID_Principal_RBG-COR.png

Figure 12: Close-loop control example: Reference & Output

Figures/Tecnico_ID_Principal_RBG-COR.png

Figure 13: Close-loop control example: Control action

3.2.2.1 Combined Open- and Closed-Loop Control

It is possible to combine open-loop and closed-loop control in a single control system. Figure 14 shows this configuration. In figures 15 and 16, the result of this configuration is shown for the same system and controllers used in the previous examples (figures 10 through to 13). In this example with the sinusoidal disturbance, it can be clearly seen that the combination of both control strategies improves the results compared to using just open-loop (which cannot adjust for the disturbance), or just closed-loop (which was slower).

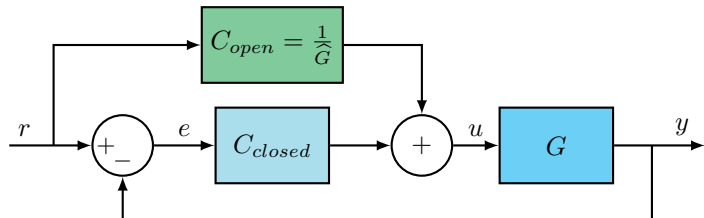


Figure 14: Combined Open- and Closed-Loop Control

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Figures/Tecnico_ID_Principal_RGB-COR.png

Figure 15: Combined open- and close-loop control example: Reference & Output

Figure 16: Combined open- and close-loop control example: Control action

3.2.2.2 PIDF Controller

Proportional-integral-derivative filtered controllers apply a control action which is the sum of three components: One proportional to the tracking error, another proportional to the integral of the error, and another proportional to a filtered derivative of the error.

A discrete PIDF controller can be formulated as in the following equation:

$$C(z) = K_p + K_i \frac{T_s}{z-1} + K_d \frac{1}{T_f + \frac{T_s}{z-1}} \quad (3.2.5)$$

Where K_p is the proportional error gain, K_i the integral error gain, K_d the derivative error gain, T_f the filter constant, and T_s the sampling time.

The proportional component outputs a control action proportional to the error. If the proportional control results in steady-state error, the integral action will grow over time and correct for it. Additional integrators may be added to correct for steady-state error. The reasoning behind the derivative term is that, if there is a large variation of the error, the control action will immediately increase to counter it, rather than waiting for a large error (where the proportional action will kick in), or, even worse, for an accumulation of the error (where the integral action will kick in).

3.2.2.3 Full State Feedback Control

In order to introduce the concept of state feedback control we'll start with a simple example. Consider the system in figure 17, which we'll attempt to stabilise using output feedback control with a proportional gain k (shown in figure 18):

$$F(t) = k_1 \cdot (x_{ref}(t) - x(t)) \quad (3.2.6)$$

Where the transfer function relating output to the reference is:

$$\frac{x(t)}{x_{ref}(t)} = \frac{k_1}{s^2 + k_1} \quad (3.2.7)$$

From which we find that the poles are on the imaginary axis ($s = \pm\sqrt{k_1}i$), the system will always oscillate, and it is not possible to stabilise the system using just output feedback control.

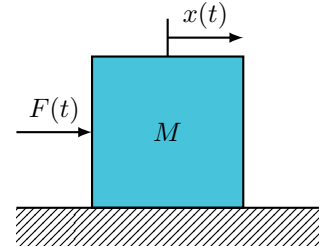


Figure 17: Mass with a force acting on it. The force is the manipulated input, and the mass position is the output.

If, instead of using just the system output (position $x(t)$) as feedback, we use an additional state (in this case the velocity $\dot{x}$, shown in figure 19) for computing the control action, we have the following control law:

$$F(t) = k_1 \cdot [x_{ref}(t) - x(t)] - k_2 \dot{x}(t) \quad (3.2.8)$$

And the closed-loop transfer function:

$$\frac{x(t)}{x_{ref}(t)} = \frac{k_1}{s^2 + k_2 s + k_1} \quad (3.2.9)$$

From which we can now see that the gains k_1 and k_2 may be chosen freely to place the closed-loop poles, p_1 and p_2 , anywhere in the s-plane:

$$\begin{aligned} s^2 + k_2 s + k_1 &= (s - p_1)(s - p_2) \implies \\ \implies k_1 &= p_1 p_2 \wedge k_2 = -p_1 - p_2 \end{aligned} \quad (3.2.10)$$

3.2.2.3.a. Optimal Control

Optimal control is precisely about designing the optimal state feedback controller gains. This is done by defining some performance index which is a function of the control action:

$$J(x(t), u(t)) = \int_{t_0}^T g(x(t), u(t)) dt \quad (3.2.11)$$

Where $x(t)$ is the states, and $u(t)$ the control action, of a given LTI system:

$$\dot{x}(t) = f(x(t), u(t)) = Ax(t) + Bu(t) \quad (3.2.12)$$

Usual performance indexes have a quadratic form to guarantee mathematical tractability. In section 4.2.2, we'll want to find the optimal gains k for reference tracking with LQR servo controller. An advantage of the linear quadratic optimal control method, over a pole-placement method, is that it provides a systematic way of computing the state feedback control gain matrix, guaranteeing a viable solution in terms of control action feasibility. To ensure the controller minimizes the tracking error we'll define the performance index (equation 3.2.13), by specifying matrices Q and R , such that the minimization of the index mostly minimizes the state, or states, which somehow represent tracking error. In the case of the LQI controller in section 4.2.2, the system will be augmented with a stated which is the integral of the tracking error, and this is the state that will be weighted by matrix Q :

$$J(x(t), u(t)) = \int_0^t x^\top Q x + u^\top R u dt \quad (3.2.13)$$

The first term in the integrand affects state response, while the second affects control action.

Notice how the performance index integral in equation 3.2.12 is defined for a finite time. The solution for the finite time control problem is optimal, however, it is a time-varying gain vector $k(t)$:

$$k(t) = R^{-1} B^\top P(t) \quad (3.2.14)$$

Which is computed by first solving $P(t)$ from the matrix differential Riccati equation:

$$-\dot{P}(t) = A^\top P(t) + P(t)A - P(t)BR^{-1}B^\top P(t) + Q \quad (3.2.15)$$

Alternatively, for the sake of simplicity, the performance index is defined for infinite time, and, in this case, as $t \rightarrow \infty$, the optimal control solution becomes the steady-state solution, with the gain vector k and matrix P becoming constant, making computation much faster and easier:

$$0 = A^\top P + PA - PBR^{-1}B^\top P + Q \quad (3.2.16)$$

$$k = R^{-1} B^\top P \quad (3.2.17)$$

This solution is suboptimal, however.

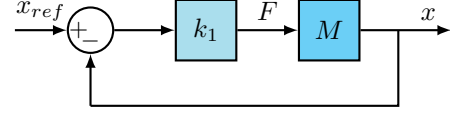


Figure 18: System from figure 17 with output feedback

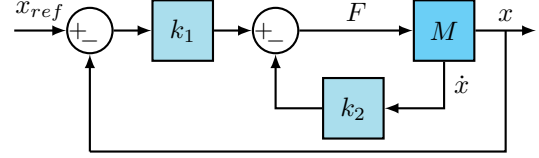


Figure 19: System from figure 17 with state feedback, where the states are position and velocity of the mass

3.2.2.3.b. Controllability

For the existence of an optimal control vector, a system must be completely state controllable. A system is said to be state controllable if it is possible to construct a finite control action that will transfer an initial state to any final state in a finite time interval. If every state is controllable, then the system is said to be completely state controllable [Sá da Costa, 2012].

It is important to note that controllability is not a property of the system, but, instead, a property of the state space representation, and is affected only by the state and input matrices, A and B , respectively [Botto, 2021].

3.2.2.3.c. Observability

In order to implement state feedback control one must, obviously, know the states. The concept of observability is useful in solving the problem of reconstructing unmeasurable state variables from measurable variables in the minimum possible length of time. A system is said to be completely observable if every state can be determined from the observation of the system output over a finite time interval. The system is, therefore, completely observable if every transition of the state eventually affects every element of the output vector [Sá da Costa, 2012].

It is important to note that observability, like controllability, is not a property of the system, but a property of the state space representation, and is affected only by the state and output matrices, A and C , respectively [Botto, 2021].

3.2.2.3.d. State Observers

A state observer estimates system states from measurements of inputs and outputs of a system. This is done such that state feedback control can be applied to systems with unmeasurable states. In section 4.2.2.4 a stationary Kalman filter (isn't recursively updated) is implemented to estimate the systems states, mixing a priori information from the identified system model with the measured output to compute an estimate of the states.

For a given plant:

$$\dot{x} = Ax + Bu + w \quad (3.2.18)$$

$$y = Cx + v \quad (3.2.19)$$

Where $w(t)$ and $v(t)$ are Gaussian white noise processes with covariance matrices:

$$Q_n = E[ww^\top] \quad (3.2.20)$$

$$R_n = E[vv^\top] \quad (3.2.21)$$

The Kalman estimator:

$$\hat{\dot{x}} = A\hat{x} + Bu + L_K(y - C\hat{x}) \quad (3.2.22)$$

$$\hat{y} = C\hat{x} \quad (3.2.23)$$

Computes state estimate $\hat{x}(t)$ that minimize steady-state error covariance:

$$P = \lim_{t \rightarrow \infty} E((x - \hat{x})(x - \hat{x})^\top) \quad (3.2.24)$$

Where the filter gain L_K is given by:

$$L_K = PC^\top R_n^{-1} \quad (3.2.25)$$

Where P is the solution of the algebraic Riccati equation:

$$AP + PA^\top + BQ_nB^\top - PC^\top R_n^{-1}CP = 0 \quad (3.2.26)$$

Matrices Q_n and R_n can be used to tune the observer, biasing the states estimates towards what our model of system expects (done when the measurements are unreliable), or what the sensors are reading (done when the model isn't good).

3.3 Identification in Closed-Loop Systems

Many systems operate under feedback control due to required safety of operation (as is the case of LNEC's shaking tables), or to unstable behaviour of the plant meaning that experiments can only be performed under presence of a stabilizing controller. Biological and economic systems operate only under closed-loop conditions, and it is not even possible to remove the feedback loop. On occasion, performing experiments in closed loop may be preferable to operating in open loop, such as a situation in which the objective of the identification is to design a better controller, then, the dynamics of the plant in closed loop with the original controller may be more relevant than the open loop for designing an improved controller [Van den Hof, 2020].

In section 3.3.2 it will be shown that the identification of systems working in closed loop can be problematic under certain conditions, with a specific focus on nonparametric modelling by spectral analysis, as this is the method currently used at LNEC when synthesizing adapted drivers for the shaking tables. In section 3.3.3 a method for circumventing the issues introduced by the feedback loop will be presented. These segments are specially influenced by the works of Van den Hof [2020] and Söderström and Stoica [1989],

3.3.1 Closed Loop System Setup

In this section we'll consider the closed loop system represented in figure 20, and restrict ourselves to the case of SISO systems (single-input single-output):

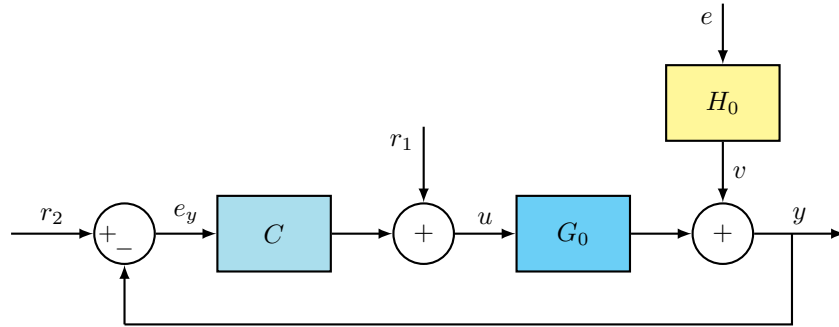


Figure 20: Closed-loop System

For the system in figure 20, the following equations are derived:

$$y(t) = G_0 \cdot u(t) + H_0 \cdot e(t) \quad (3.3.1)$$

$$u(t) = r_1(t) + C \cdot r_2(t) - C \cdot y(t) \quad (3.3.2)$$

In the whole of section 3.3, for the examples presented, and, unless stated otherwise, we'll consider the following conditions:

- $r_1(t)$, $r_2(t)$ (referred to as reference/set point signals), $e(t)$ (referred to as disturbance or noise) are Gaussian white noise data sequences generated with MATLAB's *wgn* function, with power set to 1
- $C(z)$ is the system controller, which will vary between examples
- $G_0(z) = \frac{1}{1-1.6z^{-1}+0.89z^{-2}}$ is the plant
- $H_0(z) = \frac{1-1.56z^{-1}+1.045z^{-2}-0.3338z^{-3}}{1-2.35z^{-1}+2.09z^{-2}-0.6675z^{-3}}$ is the noise model

We'll start by transforming the system equations. The reasons why this is of interest we'll become apparent soon. Firstly, for simplicity's sake, we'll substitute $e(t)$ in equation (3.3.1):

$$v(t) = H_0(z) \cdot e(t) \implies y(t) = G_0 \cdot u(t) + v(t) \quad (3.3.3)$$

And introducing the sensitivity transfer function:

$$S_0 = \frac{1}{1 + G_0 \cdot C} \quad (3.3.4)$$

And rewriting equations (3.3.1) and (3.3.2) using relations (3.3.3) and (3.3.4), we get:

$$\begin{aligned} y(t) &= S_0 G_0 \cdot r_1(t) + S_0 G_0 C \cdot r_2(t) + S_0 \cdot v(t) \\ &= H_{yr_1} \cdot r_1(t) + H_{yr_2} \cdot r_2(t) + H_{yv} \cdot v(t) \end{aligned} \quad (3.3.5)$$

$$\begin{aligned} u(t) &= S_0 \cdot r_1(t) + S_0 C \cdot r_2(t) - S_0 C \cdot v(t) \\ &= H_{ur_1} \cdot r_1(t) + H_{ur_2} \cdot r_2(t) + H_{uv} \cdot v(t) \end{aligned} \quad (3.3.6)$$

The block diagram resulting from these reworked equations is shown in Figure 21:

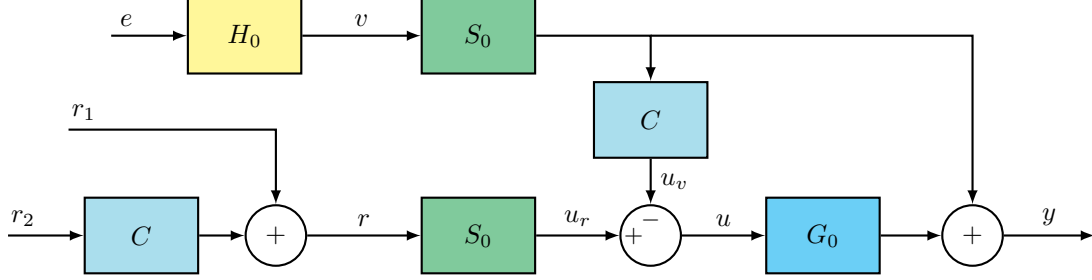


Figure 21: The closed-loop system is reformulated into pseudo-open loop form by reworking the equations. The control action, u , is now decomposed in two: the noise free control action, u_r (resulting from the reference, r) and the corrupted control action, u_v (resulting from disturbance on the output being fed back to the controller).

3.3.2 Nonparametric Modelling by Spectral Analysis

3.3.2.1 Why it works in open-loop

For our system, taking the spectral density Φ_{yu} , we get:

$$\Phi_{yu}(\omega) = G_0(e^{i\omega}) \Phi_u(\omega) + H_0(e^{i\omega}) \Phi_{eu}(\omega) \quad (3.3.7)$$

If we had an open loop system as in figure 22, where the output y (which contains the disturbance e) was not being fed back into the plant, then, given the fact that e and u are uncorrelated (meaning $\Phi_{eu}(\omega) = 0$), our plant can be estimated directly by estimating the spectral densities:

$$\hat{G}(e^{i\omega}) = \frac{\hat{\Phi}_{yu}(\omega)}{\hat{\Phi}_u(\omega)} \quad (3.3.8)$$

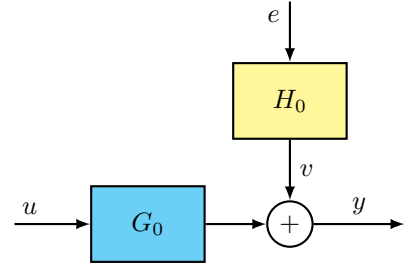


Figure 22: Open-loop System

3.3.2.2 The problem in closed-loop

Applying the same technique for our closed loop system (figure 20), from equations (3.3.5) and (3.3.6), we find:

$$\begin{aligned} \Phi_u(\omega) &= |H_{ur_1}|^2 \Phi_{r_1} + |H_{ur_2}|^2 \Phi_{r_2} + |H_{uv}|^2 \Phi_v \\ &= |S_0|^2 \Phi_{r_1} + |S_0 C|^2 \Phi_{r_2} + |-S_0 C|^2 \Phi_v \\ &= |S_0|^2 [\Phi_{r_1} + |C|^2 \Phi_{r_2} + |C|^2 \Phi_v] \end{aligned} \quad (3.3.9)$$

$$\begin{aligned} \Phi_{yu}(\omega) &= H_{yr_1} H_{ur_1}^* \Phi_{r_1} + H_{yr_2} H_{ur_2}^* \Phi_{r_2} + H_{yv} H_{uv}^* \Phi_v \\ &= S_0 G_0 \cdot S_0^* \Phi_{r_1} + S_0 G_0 C \cdot (S_0 C)^* \Phi_{r_2} + S_0 \cdot (-S_0 C)^* \Phi_v \\ &= |S_0|^2 [G_0 \Phi_{r_1} + G_0 |C|^2 \Phi_{r_2} - C^* \Phi_v] \end{aligned} \quad (3.3.10)$$

$$\hat{G}(e^{i\omega}) = \frac{\Phi_{yu}(\omega)}{\Phi_u(\omega)} = \frac{G_0 \Phi_{r_1} + G_0 |C|^2 \Phi_{r_2} - C^* \Phi_v}{\Phi_{r_1} + |C|^2 \Phi_{r_2} + |C|^2 \Phi_v} \quad (3.3.11)$$

From equation (3.3.11) we can verify that, if $C(z) = 0$, then we simply have the open-loop case as in equation (3.3.8):

$$C(z) = 0 \implies \hat{G}(e^{i\omega}) = \frac{G_0 \Phi_{r_1} + 0 - 0}{\Phi_{r_1} + 0 + 0} = \frac{G_0(e^{i\omega}) \Phi_{r_1}(\omega)}{\Phi_{r_1}(\omega)} = G_0(e^{i\omega}) \quad (3.3.12)$$

What's more interesting, though, is the closed-loop scenario, so we'll consider $C(z) \neq 0$. From equation (3.3.11) we can verify that, if the disturbance v is very small ($\Phi_v(\omega) \rightarrow 0$), the identification of the plant's true dynamics is possible:

$$\Phi_v(\omega) \rightarrow 0 \implies \hat{G}(e^{i\omega}) = \frac{G_0 \Phi_{r_1} + G_0 |C|^2 \Phi_{r_2} - 0}{\Phi_{r_1} + |C|^2 \Phi_{r_2} + 0} = \frac{G_0 \cdot [\Phi_{r_1} + |C|^2 \Phi_{r_2}]}{\Phi_{r_1} + |C|^2 \Phi_{r_2}} = G_0(e^{i\omega}) \quad (3.3.13)$$

But if the disturbance v is very large compared to the reference signals r_1 and r_2 , then it results:

$$\frac{\Phi_{r_1}(\omega)}{\Phi_v(\omega)} \rightarrow 0, \frac{\Phi_{r_2}(\omega)}{\Phi_v(\omega)} \rightarrow 0 \implies \hat{G}(e^{i\omega}) = \frac{0 + 0 - C^*}{0 + 0 + |C|^2} = \frac{-C^*}{C \cdot C^*} = \frac{-1}{C(e^{i\omega})} \quad (3.3.14)$$

So it's clear that the spectral analysis method in closed-loop always results in biased estimates of the plant, with the identified model $\hat{G}$ being somewhere between G_0 and $\frac{-1}{C}$, as weighed by the spectral density of each signal and the controller.

3.3.2.3 Remark on parametric modelling

As to complete this section, I want to touch on the subject of parametric models. As demonstrated in Van den Hof [2020] parametric modelling by direct identification can result in consistent identification of the plant, but only if one is willing to have full order models, and while simultaneously modelling the noise accurately, also while using sufficiently exciting external signal r or with noise when having a sufficiently high order controller, or one that switches settings during the experiment. Other methods require complex parametrized models, more steps, solving additional equations, etc. All these conditions often make those approaches undesirable, however, the parametric identification method introduced in section 3.3.3 is able to address the closed-loop identification issues while being of simple implementation.

3.3.3 Two-Stage Method for Closed Loop Identification

3.3.3.1 Introduction

The two-stage method is introduced in Van den Hof and Schrama (1993). It's of special interest compared to others parametric modelling approaches since it deals with the consistency problem of identifying G_0 without having to accurately model H_0 , but also due to its simplicity and flexibility. It does not impose restrictions on the controller, which can be of any order, nonlinear, of the variable structure (as in sliding mode control), etc., nor does it require prior knowledge of its dynamics. It also does not require complex parametrized models or solving additional equations when reduced order models are desired.

The simplicity results from the fact that the procedure comes down to two standard open-loop identification steps. This approach works on the principle that, by decomposing the control action, u , in two components: the noise free control action, u_r which results from the reference, r ; and the corrupted control action, u_v which results from the disturbance v on the output being fed back to the control.

$$u(t) = u_r + u_v \quad (3.3.15)$$

$$= S_0 [r_1(t) + Cr_2(t)] - S_0 Cv(t) \quad (3.3.16)$$

It is then possible to estimate G_0 , not from u (which is corrupted), but from y and u_r , since u_r is uncorrelated to v and u_v :

$$\begin{aligned} y(t) &= G_0 u(t) + v(t) \\ &= G_0 [u_r(t) + u_v(t)] + v(t) \\ &= G_0 u_r(t) + [G_0 u_v(t) + v(t)] \end{aligned} \quad (3.3.17)$$

3.3.3.2 The Method

3.3.3.2.a. Estimating the sensitivity function

If we mind equation 3.3.6, condensing the external reference signals in a single variable:

$$r(t) = r_1(t) + C(z) \cdot r_2(t) \quad (3.3.18)$$

And, rewriting the equation, we get:

$$u(t) = S_0 \cdot r(t) - S_0 C \cdot v(t) \quad (3.3.19)$$

Now the reason for formulating the problem using the sensitivity function becomes apparent, due to the fact that $r(t)$ and $v(t)$ are uncorrelated, S_0 can be estimated in a usual open-loop way from the measured $u(t)$ signal, and the known reference signal $r(t)$. This is the first step:

$$\widehat{S}(z) = \frac{u(t)}{r(t)} \quad (3.3.20)$$

3.3.3.2.b. Reconstructing the noise-free control action

From the known reference signal $r(t)$, the identified sensitivity function can be used to compute the noise-free control action $\widehat{u}_r(t)$:

$$\widehat{u}_r(t) = \widehat{S}(z) \cdot r(t) \quad (3.3.21)$$

Since the sole purpose of the sensitivity function is computing a noise-free control action, to this end, the model can be of arbitrarily high order, but restricting variance, and thus reducing the number of parameters, may still be of interest.

3.3.3.2.c. Estimating the plant

From the computed noise-free control action, $\widehat{u}_r$, and measured system output y , it is now possible to estimate the plant:

$$\widehat{G}(z) = \frac{y(t)}{\widehat{u}_r(t)} \quad (3.3.22)$$

3.3.4 Example: Comparing Closed-Loop Identification Methods

The following examples display the identification results obtained with the methods discussed in sections 3.3.2 and 3.3.3. To illustrate the effects of the controller in the identification results, in the first example the controller was set to a proportional gain, and in the second, to a weighted sum of two delays. For each controller, two scenarios are shown: One where the ratio of disturbance to external excitation (reference signal) is moderate, and another where the disturbance is more severe.

For this section, consider the following:

- A data sequence containing 32768 points was used
- $r_1(t) = 0$, where the tracking error ($e_y = r_2 - y$) is what is input to the controller, as this mimics the setup at LNEC
- The data sequences $r_2(t)$ and $e_0(t)$ for the pairs of examples are the same, with the noise input $e(t)$ in each one resulting from a scaling of $e_0(t)$: $e(t) = k \cdot e_0(t)$
- In figures 23 and 24, the controller was set to $C(z) = 5$
- In figures 25 and 26, the controller was set to $C(z) = 0.5 \cdot (z^{-1} + 0.8z^{-2})$



Figure 23: Bode plot of identification results with controller $C(z) = 5$ and disturbance set as $e(t) = 0.1 \cdot e_0(t)$

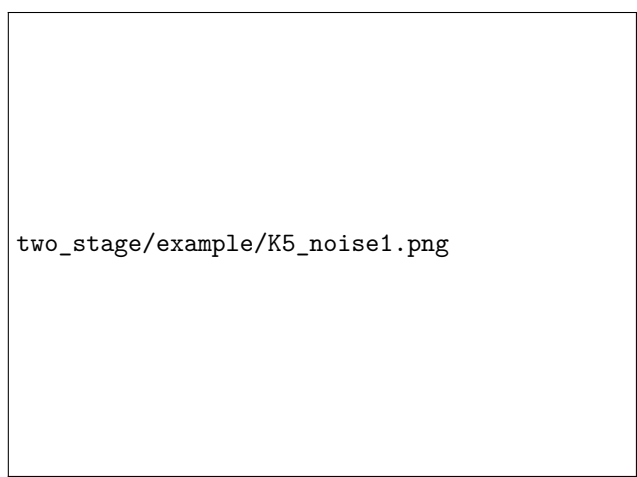


Figure 24: Bode plot of identification results with controller $C(z) = 5$ and disturbance set as $e(t) = 1 \cdot e_0(t)$

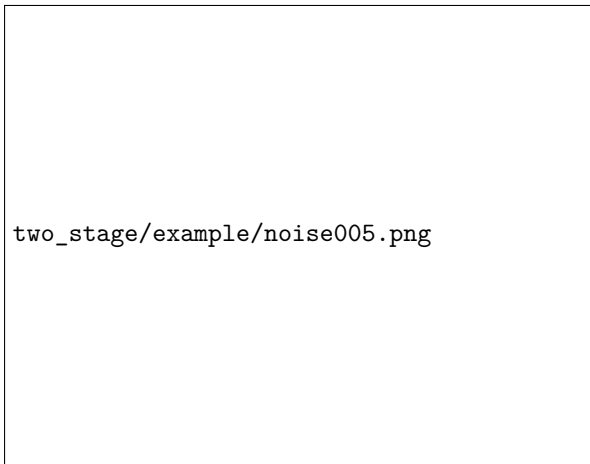


Figure 25: Bode plot of identification results with controller $C(z) = 0.5 \cdot (z^{-1} + 0.8z^{-2})$ and disturbance set as $e(t) = 0.05 \cdot e_0(t)$

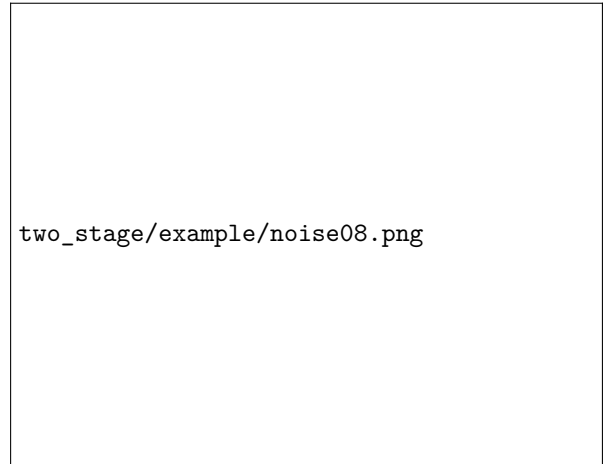


Figure 26: Bode plot of identification results with controller $C(z) = 0.5 \cdot (z^{-1} + 0.8z^{-2})$ and disturbance set as $e(t) = 0.8 \cdot e_0(t)$

The figure on the left shows the results of the identification with some noise, but not too severe, and here we can see that the results of the non-parametric models are very sensitive to windowing. On the one hand, a smaller window (pink dots) results in a model that is less sensitive to variations in the data, but with a high bias. A larger window (light blue dots) captures the dynamics of the system well, but suffers from high variance. As for parametric methods, we see that the two-stage method achieves excellent identification (green line, completely superimposed on the ideal model, in red), while the direct method fails. In the figure on the right, we see an example with extreme noise, such that the models obtained by non-parametric methods and the direct method are approaching the inverse of the controller. Here, the two-stage method shines by giving still acceptable results, despite everything.

4 Evaluating Controller Performance in Simulation Environment

In this section a model of LNEC's uniaxial shaking table (section 4.1) is used to evaluate the performance of three control strategies: the OLI method currently in use at LNEC (in section 4.2.3), tuned PID-F controllers (in section 4.2.1), and optimal controllers (in section 4.2.2).

4.1 A Model of LNEC's Uniaxial Seismic Table

4.1.1 Previous Works

In Oliveira [2015] a complete schematic representation of the ST1D, with a coupled 2DOF system, and instrumentation, can be found (figure 1).

In Tekeste [2021] a model for the ST1D with a coupled 1DOF system is devised from linearized analytical models for each of its components (controller, servo valve, hydraulic actuator, platen mass, and total damping).

Modelling/Gidewon_block_diagram.png

Figure 27: Schematic diagram of a shaking table system with payload rigidly attached [Tekeste, 2021]

In the previously mentioned publication, the parameters in table 1 were determined for the model in figure 27.

It's important to note that the servo valve and hydraulic actuators have since been updated. Until a new identification of the updated system has been performed, the parameters determined in Tekeste [2021] are the ones which will be used in all simulation studies.

Table 1: ST1D model parameters [Tekeste, 2021]

Symbol	Description	Value (Units)
k_p	Proportional gain of the controller	1.2993 V/cm
k_{sv}	Servo-valve gain	
τ_{sv}	Time-delay parameter of the servo-valve transfer function	0.0246 s
$k_{sv}k_q$	Valve flow gain	1934.50 cm ³ /s/V
k_c	Valve pressure-flow gain	
C_l	Total leakage coefficient of the piston	
k_{Pl}	$= k_c + C_l$	$1.67401 \cdot 10^{-7}$ m ³ /s/kPa
A	Area of the fluid under compression in the actuator	0.012456 m ²
V_t	Total volume of the fluid under compression in the actuator	0.002659 m ³
β_e	Effective bulk modulus of the system	193716.28 kPa
K_h	Oil-column frequency of the actuator	4531.79 kPa/m
m_p	Mass of the platen	1.9751 ton
c_t	Combined damping force of the actuator and the platen	5.7800 kNs/m
k_{sp}	Stiffness of the SDOF structure	-
c_{sp}	Damping of the SDOF structure	-
m_T	Total mass considering the payload (shaking table)	-

A *Simulink* model was built in order to simulate the ST1D dynamics, however, its implementation resulted in a build up of numerical errors leading to overflow, making this modelling approach unviable.

4.1.2 Mathematical Model for LNEC Uniaxial Seismic Table with a Coupled 2DoF System

Given the previous results, an alternative approach was found in order to simulate the system, which consisted on obtaining the transfer function relating the desired platen position, x_{ref} , as input, and the force applied to the platen, F_P , as output.

Modelling/ST1D_block_diagram.png

Figure 28: Block diagram of the ST1D

From the block diagram in figure 28, the following equations were derived:

$$\begin{cases} G_C(s) = \frac{i_{sv}}{e} = \frac{i_{sv}}{x_{ref} - x_T} \\ \frac{Q_L}{x_{ref} - x_T} = \frac{Q_L}{e} = K_q \cdot \frac{K_{sv}}{1 + \tau_{sv}s} \cdot G_C(s) \end{cases} \implies G_{csv}(s) = \frac{Q_L}{e} = \frac{k_{sv}k_q G_c}{1 + \tau_{sv}s} \quad (4.1.1)$$

$$\begin{aligned} P_L &= \frac{1}{C_l + \frac{A^2}{k_h} \cdot s} [Q_L - A \cdot s \cdot x_T - k_c \cdot P_L] \iff \\ \iff \frac{F_P}{i_{sv}} &= \frac{A \cdot \frac{k_{sv}k_q}{1 + \tau_{sv}s}}{k_{Pl} + \frac{A^2}{k_h} \cdot s + A^2 \cdot s \cdot G_{xT/F_P}} = G_{F_P/i_{sv}}(s) \quad , \text{ where } k_{Pl} = k_c + C_l \end{aligned} \quad (4.1.2)$$

It's important to note that the transfer function $G_{F_P/i_{sv}}(s)$ is also a function of the model coupled to the platform (see equations (4.1.2) & 4.1.4).

To represent the dynamics of a typical structure on test (which has two vibration modes in the bandwidth of the seismic table), a 2 degree of freedom mass-spring-damper system is coupled to the platen, as depicted in figure 29:

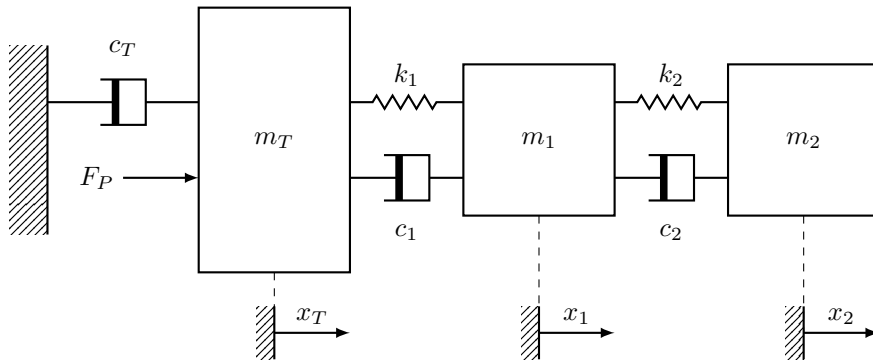


Figure 29: Schematic representation of the 2 degree of freedom mass-spring-damper system coupled to the platen

For the system in figure 29 the following equations of motion are obtained (system of equations (4.1.3)):

$$\begin{cases} m_T \cdot \ddot{x}_T + c_T \cdot \dot{x}_T + c_1(\dot{x}_T - \dot{x}_1) + k_1(x_T - x_1) = F_p \\ m_1 \cdot \ddot{x}_1 + c_1(\dot{x}_1 - \dot{x}_T) + c_2(\dot{x}_1 - \dot{x}_2) + k_1(x_1 - x_T) + k_2(x_1 - x_2) = 0 \\ m_2 \cdot \ddot{x}_2 + c_2(\dot{x}_2 - \dot{x}_1) + k_2(x_2 - x_1) = 0 \end{cases} \quad (4.1.3)$$

$$\Leftrightarrow \begin{cases} m_T \ddot{x}_T + (c_T + c_1)\dot{x}_T + k_1 x_T - c_1 \dot{x}_1 - k_1 x_1 = F_p \\ m_1 \ddot{x}_1 + (c_1 + c_2)\dot{x}_1 + (k_1 + k_2)x_1 - c_1 \dot{x}_T - c_2 \dot{x}_2 - k_1 x_T - k_2 x_2 = 0 \\ m_2 \ddot{x}_2 + c_2 \dot{x}_2 + k_2 x_2 - c_2 \dot{x}_1 - k_2 x_1 = 0 \end{cases} \Leftrightarrow$$

$$\Leftrightarrow \begin{bmatrix} m_T & 0 & 0 \\ 0 & m_1 & 0 \\ 0 & 0 & m_2 \end{bmatrix} \begin{bmatrix} \ddot{x}_T \\ \ddot{x}_1 \\ \ddot{x}_2 \end{bmatrix} + \begin{bmatrix} c_T + c_1 & -c_1 & 0 \\ -c_1 & c_1 + c_2 & -c_2 \\ 0 & -c_2 & c_2 \end{bmatrix} \begin{bmatrix} \dot{x}_T \\ \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} + \begin{bmatrix} k_1 & -k_1 & 0 \\ -k_1 & k_1 + k_2 & -k_2 \\ 0 & -k_2 & k_2 \end{bmatrix} \begin{bmatrix} x_T \\ x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} F_p \\ 0 \\ 0 \end{bmatrix}$$

From the system of equations (4.1.3), the following transfer functions relating the displacements of each mass to the displacements of an adjacent mass, or the force applied to the platen, are obtained:

$$\frac{x_T}{F_p} = \frac{G_1 G_2 - G_{21}^2}{G_T G_1 G_2 - G_T G_{21}^2 - G_2 G_{T1}^2} = G_{xT/F_p}(s) \quad (4.1.4) \quad G_T(s) = m_T \cdot s^2 + (c_T + c_1) \cdot s + k_1 \quad (4.1.7)$$

$$\frac{x_1}{x_T} = \frac{G_{T1} G_2}{G_1 G_2 - G_{21}^2} = G_{x1/xT}(s) \quad (4.1.5) \quad G_1(s) = m_1 \cdot s^2 + (c_1 + c_2) \cdot s + k_1 + k_2 \quad (4.1.8)$$

$$\frac{x_2}{x_1} = \frac{G_{21}}{G_2} = G_{x2/x1}(s) \quad (4.1.6) \quad G_2(s) = m_2 \cdot s^2 + c_2 \cdot s + k_2 \quad (4.1.9)$$

$$G_{T1}(s) = c_1 \cdot s + k_1 \quad (4.1.10)$$

$$G_{21}(s) = c_2 \cdot s + k_2 \quad (4.1.11)$$

$$G_{Fp/x_{ref}}(s) = \frac{G_{csv}}{\frac{k_{pl}}{A} + \frac{A}{k_h} s + G_{xT/Fp}(G_{csv} + A \cdot s)} \quad (4.1.12)$$

$$G_{xT/x_{ref}}(s) = G_{Fp/x_{ref}} \cdot G_{xT/Fp} \quad (4.1.13)$$

The mass m_i , stiffness k_i , and damping c_i , of each element will be determined such that the system's natural frequencies f_i and damping ratios ξ_i are similar to the structures in test. Equation (4.1.14) relates the natural frequency of the two modes of the 2DoF system to its spring and mass properties, and equation (4.1.15) relates the damping coefficient and ratio to mass and natural frequency.

$$\omega_{1,2}^2 = \frac{1}{2} \frac{m_1 k_2 + m_2 (k_1 + k_2)}{m_1 m_2} \mp \frac{1}{2} \sqrt{\left(\frac{m_1 k_2 + m_2 (k_1 + k_2)}{m_1 m_2} \right)^2 - 4 \frac{k_1 k_2}{m_1 m_2}} \quad (4.1.14)$$

$$\xi_i = \frac{c_i}{2 \cdot m_i \cdot \omega_i} \quad (4.1.15)$$

4.2 Control Strategies

4.2.1 Tuned PIDF

Taking into consideration the test procedure at LNEC (figure 2), where the identification is carried out each time the target signal is scaled up, or even additional times for various other reasons, one possible straightforward improvement would be to update the PI controller tune using this information. Another simple enhancement would be to add a filtered derivative term to the controller (making it a PIDF), since this should be of simple implementation in the existing version of the control software due to the fact that it's just an extension of the existing controller structure.

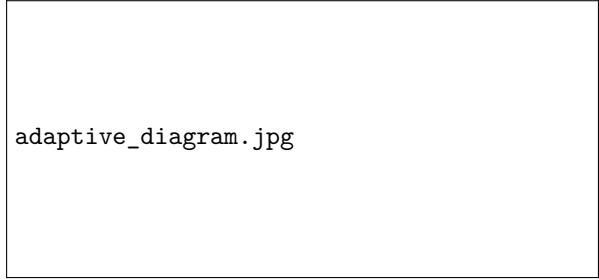


Figure 30: Adaptive control scheme

For a preliminary analysis of the mathematical model, some simulations were run using the seismic data from the 1979 Imperial Valley Earthquake, El Centro (Described in Narasimhan et al. [2006] & available here), which was scaled down to 73.51%, in order to comply with the displacement range of the seismic table. Table 2 contains the parameters for the 2DOF coupled structure (Figure 29).

System Properties	1st Mode	2nd Mode
Frequency (f_i) [Hz]	4	10
Damping Ratio (ζ_i)	0.02	0.05
Element's Properties	Element 1	Element 2
Mass (m_i) [kg]	2000	2000
Stiffness (k_i) [N/m]	$3.30 \cdot 10^5$	$1.89 \cdot 10^6$
Damping Coefficient (c_i) [N·m/s]	$1.26 \cdot 10^3$	$1.26 \cdot 10^4$

Table 2: Modal parameters for the system, and resulting stiffness and damping for each element

Initially the simulations were run with the controller set as $G_c(s) = 129.93 \text{ V/m}$, which is the tune typically used at LNEC. Afterwards, a simulation was run with $G_c(s)$ as:

$$G_c(s) = K_p + \frac{K_i}{s} + \frac{K_d \cdot s}{T_f \cdot s + 1} \quad \text{where: } K_p = 1540, K_i = 1670, K_d = 17.9, T_f = 6.96 \cdot 10^{-5} \quad (4.2.1)$$

Where the parameters K_p, K_i, K_d and T_f were obtained using MATLAB's *pidtune* function, with the tuning option '*DesignFocus*' set to '*reference-tracking*'. These results are labelled as "tuned" in figures 31 through to 39.

The code implementation can be found at the following GitHub repository location.

Sim_Res/Response_Spectra.png

Figure 31: Response spectrum for fault normal signals. The response spectrum was obtained by sampling 500 frequencies, logarithmically spaced between 0.1 and 30 Hz

In figures 34 to 37, label "MSE" refers to the respective mean squared error when using the standard controller, and "Tuned MSE" refers to the respective mean squared error when using the tuned PIDF controller.

The acceleration response spectrum mean squared error for the tuned platform is significantly reduced compared to the standard tune (Table 3), however, it exceeds the desired response spectrum for the entire frequency spectrum.

Controller	Acceleration MSE (m/s^2)	Displacement MSE (m)
Baseline	1.78	$1.94 \cdot 10^{-6}$
Tuned	0.0327	$1.88 \cdot 10^{-6}$

Table 3: Mean squared error for response spectrum of platen with ground response spectrum as baseline

Sim_Res/Bode_of_G_xT_xref.png

Figure 32: Bode plots of $G_{xT/x_{ref}}(s)$ before and after tuning

Sim_Res/Input_to_Servo.png

Figure 33: Input to servo before and after tuning



Figure 34: Accelerations before and after tuning



Figure 35: Displacements before and after tuning



Figure 36: Accelerations before and after tuning
- Detailed view of the signals covering the period
between 1.5 and 2.7 seconds



Figure 37: Displacements before and after tuning
- Detailed view of the signals covering the period
between 14.1 and 15.7 seconds

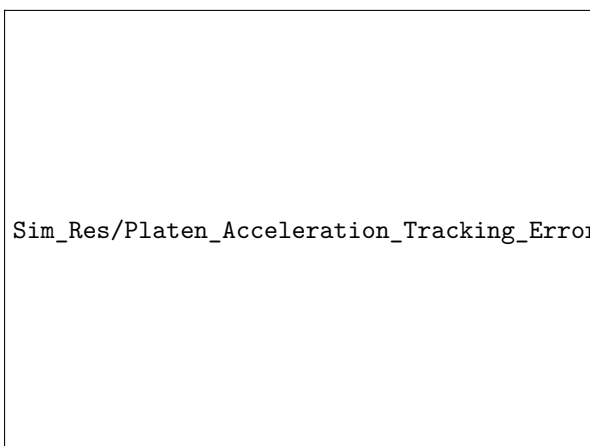


Figure 38: Acceleration tracking error before and
after tuning



Figure 39: Displacement tracking error before and
after tuning

4.2.1.1 Performance Sensitivity to Test Structure Parameters

The mass, modal frequencies, and damping of the structures coupled to the seismic table have an impact on how well it is able to track the reference. In this section, we will analyse how the seismic table with the tuned PIDF performs for different damping, modal frequencies, and mass of the test structure.

The legend of figures 40 through to 42, shows the modal properties of the structure on test, the maximum absolute values of the input to the servo-valve i_{sv} (in volts) and force applied by the actuator (in kilo newtons), and the mean square error for the respective response spectrum.

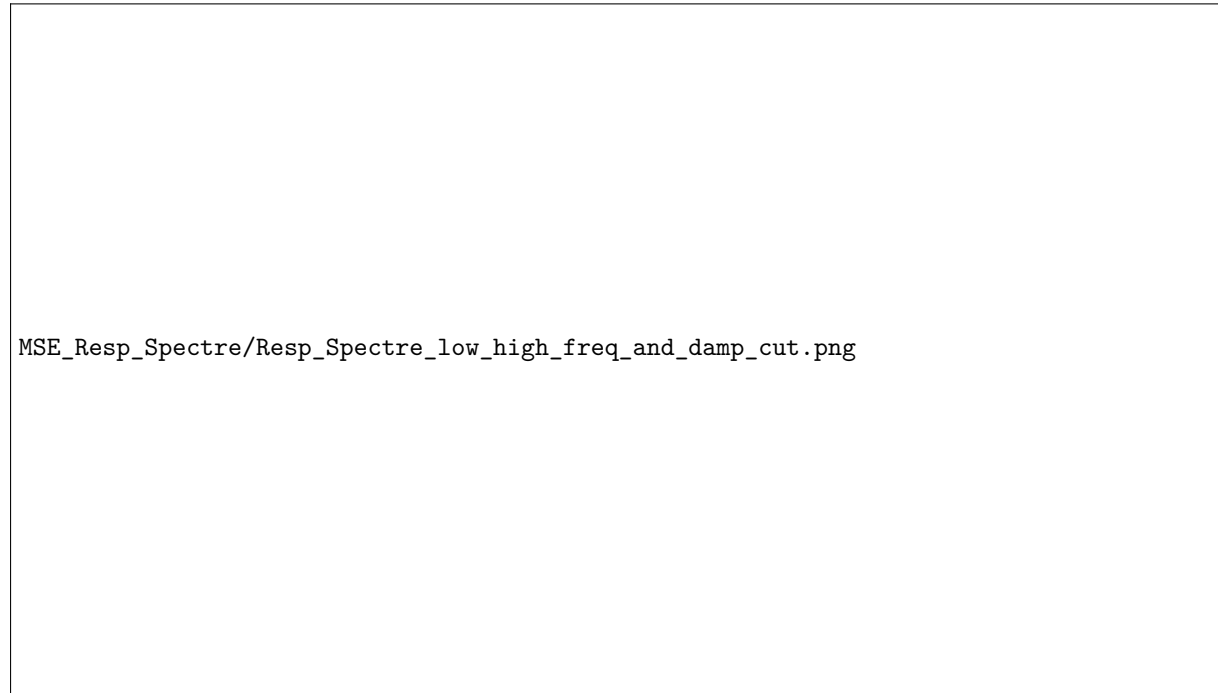


Figure 40: Response spectra for $m_1 = m_2 = 2$ ton, and different modal frequencies and damping

MSE_Resp_Spectre/Resp_Spectre_f2=10_cut.png

Figure 41: Response spectra for $m_1 = m_2 = 2$ ton, with $f_2 = 10$ Hz

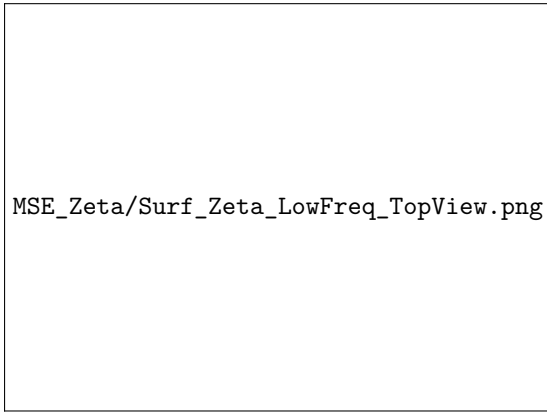
MSE_Resp_Spectre/Resp_Spectre_f1_2415_cut.png

Figure 42: Response spectra for $m_1 = m_2 = 2$ ton, and modal frequencies which are close together ($f_2 \approx 2.415 \cdot f_1$) (see figure 45)



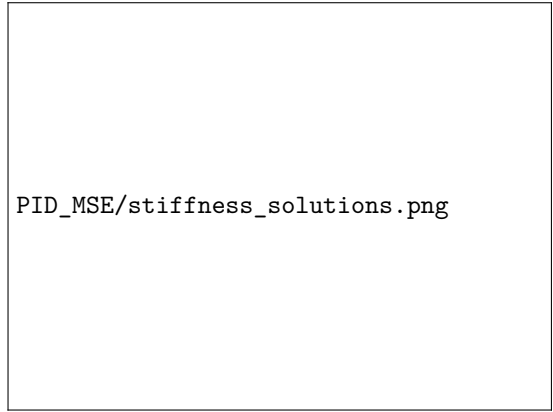
MSE_Zeta/Surf_Zeta_LowFreq_IsoView.png

Figure 43: Response Spectra MSE vs ζ_1 and ζ_2 , with $m_1 = m_2 = 2$ ton, $f_1 = 1.5Hz$, $f_2 = 6Hz$



MSE_Zeta/Surf_Zeta_LowFreq_TopView.png

Figure 44: Response Spectra MSE vs ζ_1 and ζ_2 , with $m_1 = m_2 = 2$ ton, $f_1 = 1.5Hz$, $f_2 = 6Hz$



PID_MSE/stiffness_solutions.png

Figure 45: For the 2DOF model (fig. 29), the modal frequency f_2 must be greater than $2.415 \times f_1$ (Equation (4.1.14))

MSE_Zeta/MSE vs Zeta (m_i=2.0,f_1=4.0, f_2=10.0)_old.png

MSE_Zeta/MSE vs Zeta (m_i=2.0,f_1=3.0, f_2=7.2).png

Figure 46: Response Spectra MSE vs ζ_1 and ζ_2 , with $m_1 = m_2 = 2$ ton, $f_1 = 4Hz$, $f_2 = 10Hz$. For high modal frequencies, the response spectrum MSE increases abruptly for $\zeta_2 > 0.14 - 0.8 \cdot \zeta_1$, but remains mostly unchanged otherwise.

Figure 47: Response Spectra MSE vs ζ_1 and ζ_2 , with $m_1 = m_2 = 2$ ton, $f_1 = 3Hz$, $f_2 = 7.2Hz$ (modal frequencies close together)

PID_MSE/PID_MSE_Surface_frequency_blue_red_isoview_lowDamp.png

Figure 48: Response Spectra MSE vs f_1 and f_2 , with $m_1 = m_2 = 2$ ton, $\zeta_1 = 0.02$, $\zeta_2 = 0.05$ (low damping)

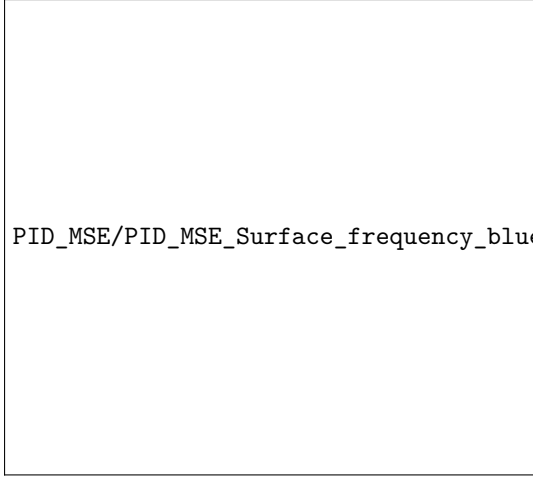


Figure 49: Response Spectra MSE vs f_1 and f_2 , with $m_1 = m_2 = 2$ ton, $\zeta_1 = 0.02$, $\zeta_2 = 0.05$ (low damping)

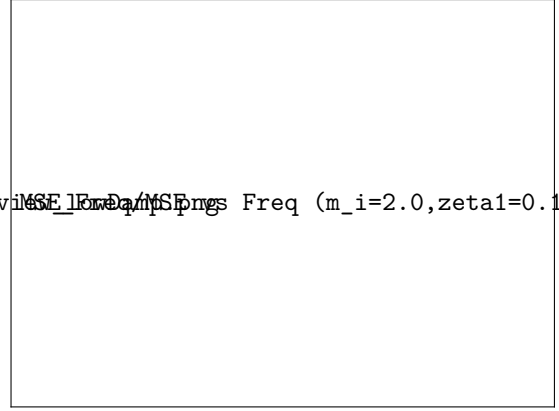


Figure 50: Response Spectra MSE vs f_1 and f_2 , with $m_1 = m_2 = 2$ ton, $\zeta_1 = 0.1$, $\zeta_2 = 0.25$ (high damping)

Structures with lower damping ($\zeta_1 = 0.02$ & $\zeta_2 = 0.05$) result in response spectra which exceeds the desired one over the entire frequency spectrum, while structures with higher damping ($\zeta_1 = 0.1$ & $\zeta_2 = 0.25$) result in response spectra which fall short of the desired one.

The effect of modal frequencies on the performance of the seismic table is unclear, and likely depends heavily on the frequency content of the seismic signal.

4.2.2 Optimal Control

4.2.2.1 State Space Model

For the seismic table with a coupled 2-DoF system, the following state space model was derived, where:

$$\dot{x}(t) = A \cdot x(t) + B \cdot u(t) \quad (4.2.2)$$

$$y(t) = C \cdot x(t) + D \cdot u(t) \quad (4.2.3)$$

With $x(t)$ being the state vector:

$$x = [Q_{sv}, F_p, x_T, x_1, x_2, \dot{x}_T, \dot{x}_1, \dot{x}_2]^T \quad (4.2.4)$$

And the control action $u(t)$, which is the signal to the servo-valve i_{sv} , the state equations are:

$$\begin{bmatrix} \dot{Q}_{sv} \\ \dot{F}_p \\ \dot{x}_T \\ \dot{x}_1 \\ \dot{x}_2 \\ \ddot{x}_T \\ \ddot{x}_1 \\ \ddot{x}_2 \end{bmatrix} = \begin{bmatrix} \frac{-1}{\tau_{sv}} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{k_h}{A} & \frac{-k_h k_{pl}}{A^2} & 0 & 0 & 0 & 0 & -k_h & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & \frac{1}{m_T} & \frac{-k_1}{m_T} & \frac{k_1}{m_T} & 0 & \frac{-c_T - c_1}{m_T} & \frac{c_1}{m_T} & 0 & 0 \\ 0 & 0 & \frac{k_1}{m_1} & \frac{-k_1 - k_2}{m_1} & \frac{k_2}{m_1} & \frac{c_1}{m_1} & \frac{-c_1 - c_2}{m_1} & \frac{c_2}{m_1} & 0 \\ 0 & 0 & 0 & \frac{k_2}{m_2} & \frac{-k_2}{m_2} & 0 & \frac{c_2}{m_2} & \frac{-c_2}{m_2} & 0 \end{bmatrix} \begin{bmatrix} Q_{sv} \\ F_p \\ x_T \\ x_1 \\ x_2 \\ \dot{x}_T \\ \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} + \begin{bmatrix} \frac{k_{sv} k_q}{\tau_{sv}} \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \cdot i_{sv} \quad (4.2.5)$$

With the system output being the platen position x_T , and without input feedthrough:

$$y(t) = C \cdot x(t) + D \cdot u(t) = x_T \implies C = [00100000], D = 0 \quad (4.2.6)$$

4.2.2.2 Evaluating Controllability & Observability

Controllability and observability matrices were built in MATLAB with *ctrb* and *obsv* commands while using variable precision arithmetic (*vpa*) to evaluate each element to ten thousand significant digits, and were verified to be of full rank.

4.2.2.3 LQR Servo Controller

For the tracking loop shown in figure 51, where *sys* is the state space model presented in section 4.2.2.1, the optimal state feedback control law which minimizes the cost function (equation (4.2.7)) was computed using MATLAB *lqi* function, for $R = 10^{-9}$ and different Q matrices.

The state weight matrix Q had every element set to zero, except the entry corresponding to the integrated tracking error x_i , which as initially set to 10^3 , which resulted in a similar bandwidth to the tuned PIDF (figure 52), and then increased to 10^9 , because this increased the bandwidth, but, unlike the PIDF, did not result in amplification at low frequencies. (Figure 53 to 62).

The code implementation can be found at the following GitHub repository location.

LQI/LQI_loop.png

Figure 51: Tracking loop with full state feedback control.

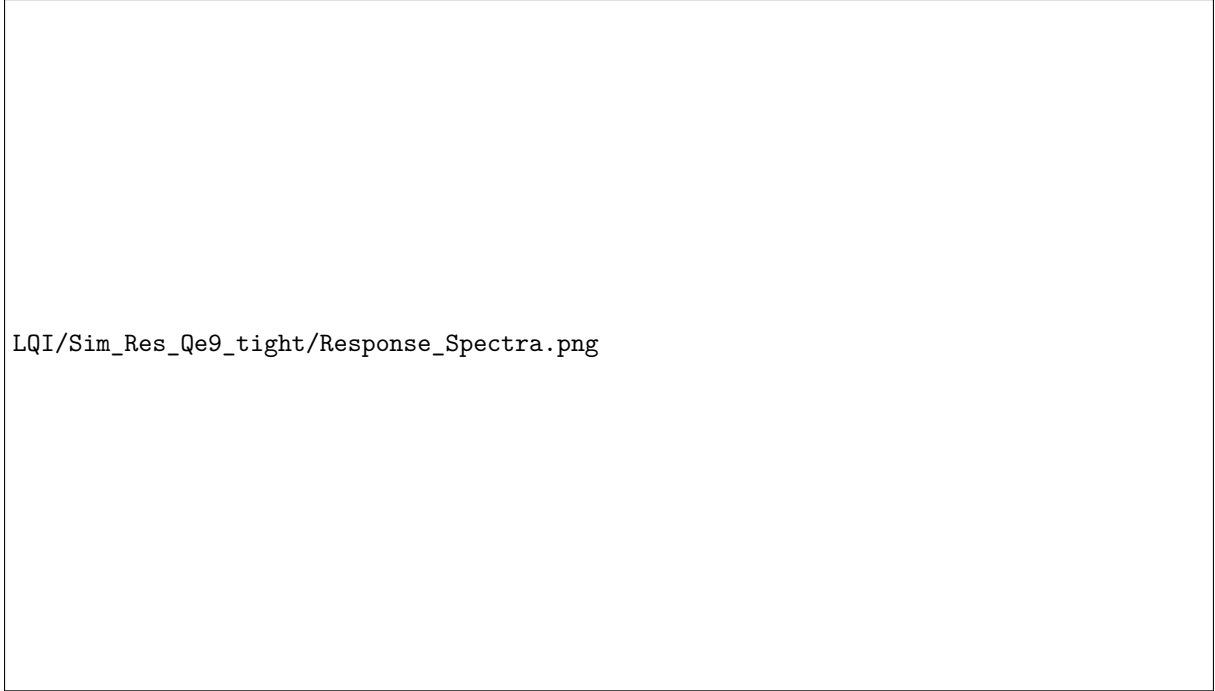
$$J(u) = \int_0^\infty (x^T(t)Qx(t) + u^T(t)Ru(t)) dt \quad (4.2.7)$$

LQI/Sim_Res_Qe3/Bode_of_G_xT_xref.png

Figure 52: Bode plots of $G_{xT/x_{ref}}(s)$ for the default controller, tuned PIDF, and optimal state-feedback controller with $Q = 10^3$ (which results in a similar bandwidth to the tuned PIDF).

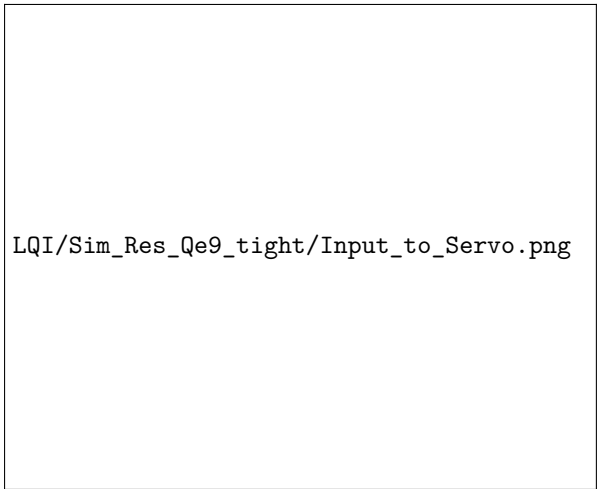
LQI/Sim_Res_Qe9/Bode_of_G_xT_xref.png

Figure 53: Bode plots of $G_{xT/x_{ref}}(s)$ for the default controller, tuned PIDF, and optimal state-feedback controller with $Q = 10^9$.



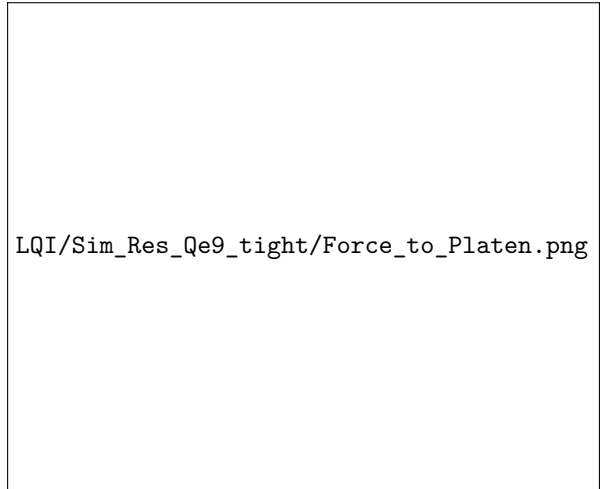
LQI/Sim_Res_Qe9_tight/Response_Spectra.png

Figure 54: Response spectrum for fault normal. The response spectrum was obtained by sampling 500 frequencies logarithmically spaced between 0.1 and 30 Hz



LQI/Sim_Res_Qe9_tight/Input_to_Servo.png

Figure 55: Input to servo



LQI/Sim_Res_Qe9_tight/Force_to_Platten.png

Figure 56: Force to Platen

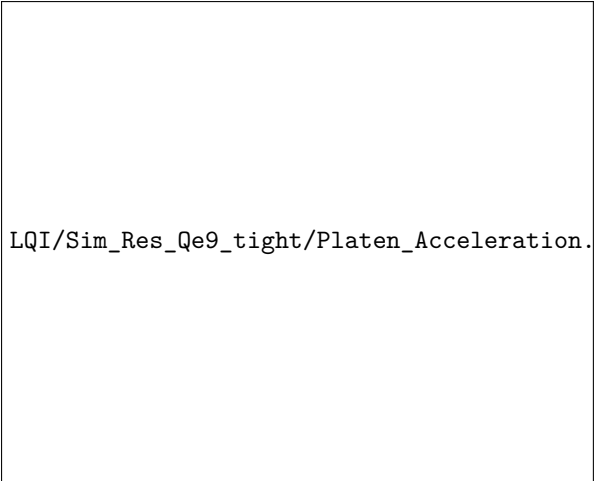


Figure 57: Accelerations

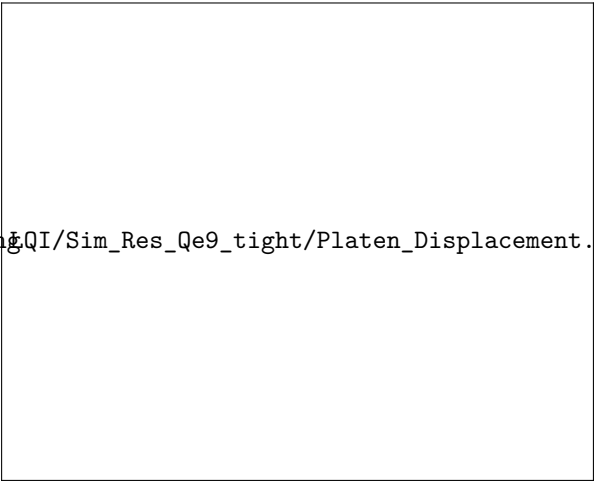


Figure 58: Displacements

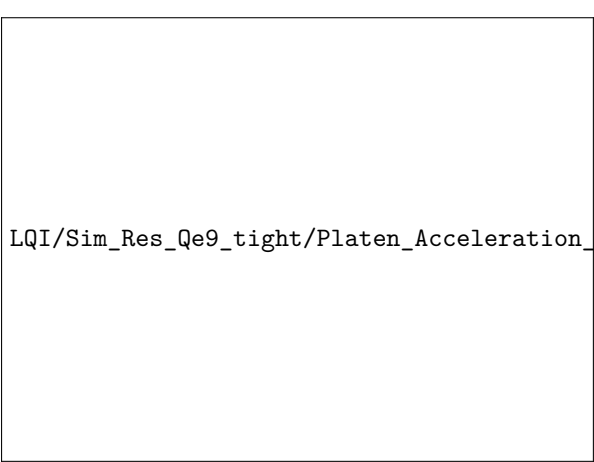


Figure 59: Accelerations - Detailed view of the signals covering the period between 1.6 and 2.8 seconds

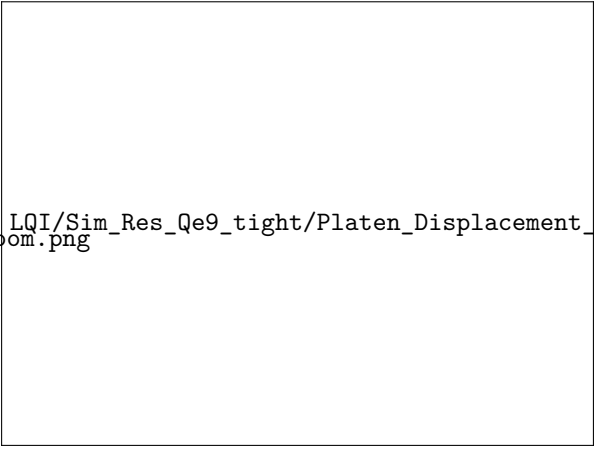


Figure 60: Displacements - Detailed view of the signals covering the period between 14.2 and 15.7 seconds

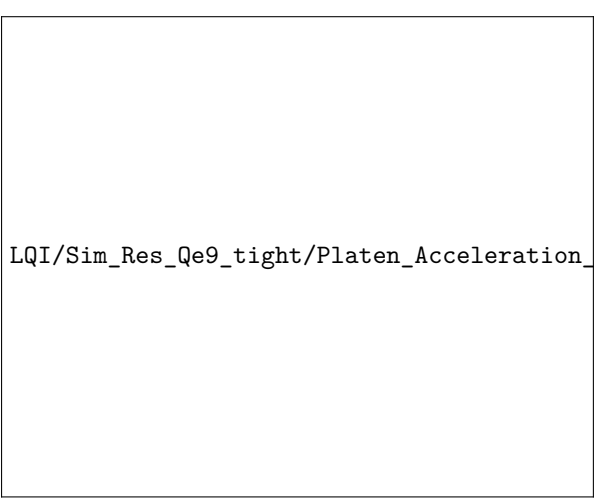


Figure 61: Acceleration tracking error



Figure 62: Displacement tracking error

4.2.2.4 LQG Servo Controller

Given how state feedback controllers requires some knowledge of the system states, either by measurement or estimation, to compute the control action, and that on the shaking platform the only measurable states are platen displacement and/or acceleration, it is clear that an implementation of an optimal controller requires also the implementation of an observer. To this end, a stationary Kalman filter was used to reconstruct the states. Implemented in MATLAB using the function *kalman* to compute the state estimator (*kest* in figure 63) from the augmented system and the process and sensor noise covariance matrices, Q_n and R_n (set as identity matrices). The state feedback control law k was computed using MATLAB *lqi* function, for $R = 10^{-9}$ and different Q matrices (The state weight matrix Q had every element set to zero, except the entry corresponding to the integrated tracking error x_i , which as initially set to 10^3).

LQG/lqgtrack1.png

Figure 63: LQG loop

LQG/LQG_loop.png

Figure 64: Controller Block Diagram

The MATLAB function *lqgtrack* was used to compute the two-degree-of-freedom LQG servo controller C (as shown in figure 63, with its internal structure shown in figure 64) by connecting the Kalman estimator *kest* and the state-feedback gain k .

The code implementation can be found at the following GitHub repository location.

LQG/Sim_Res_Qe9_tight/Response_Spectra.png

Figure 65: Response spectrum for fault normal. The response spectrum was obtained by sampling 500 frequencies logarithmically spaced between 0.1 and 30 Hz

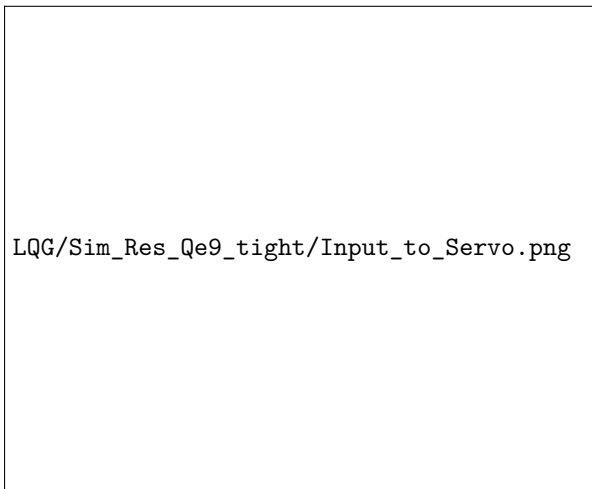


Figure 66: Input to servo

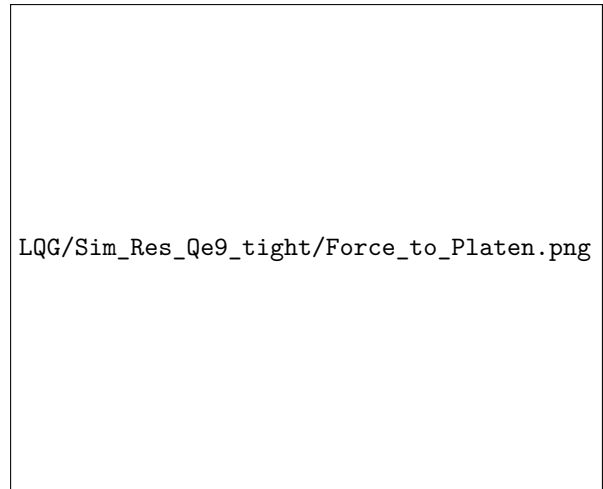


Figure 67: Force to Platen

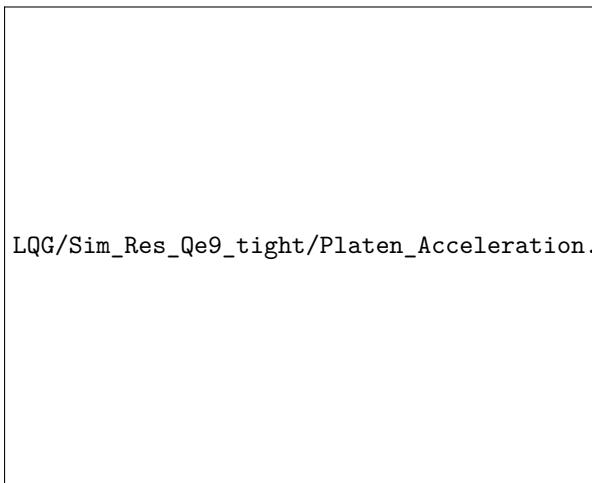


Figure 68: Accelerations

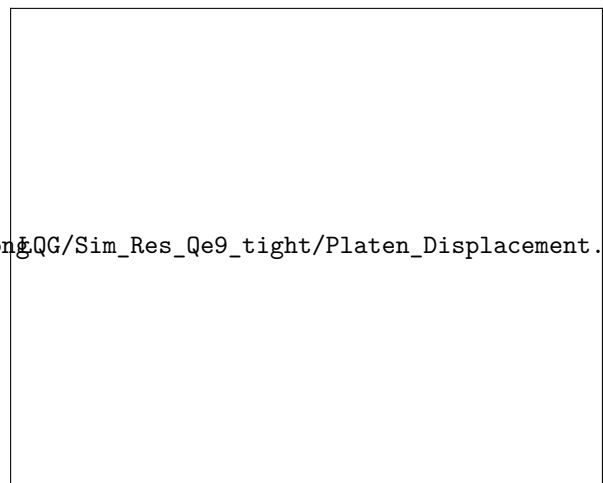


Figure 69: Displacements

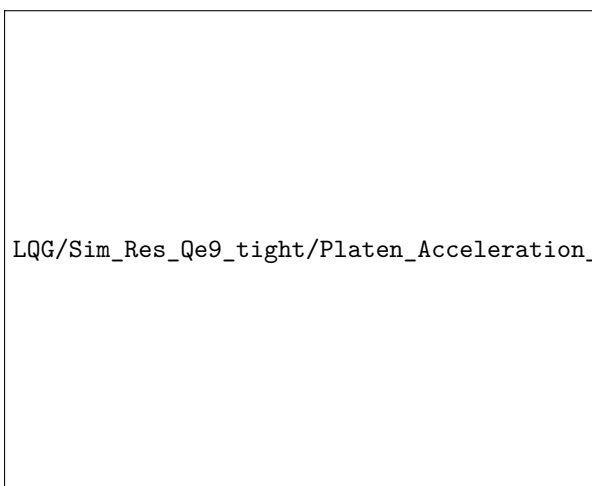


Figure 70: Acceleration tracking error



Figure 71: Displacement tracking error

4.2.3 LNEC's seismic platform driver signal adaptation process

4.2.3.1 The functioning of the driver adaptation software

As seen in the previous chapters, the use of a proportional controller for tracking the position reference signal does not produce the desired performance from the seismic table in terms of response spectrum.

An attempt to improve the seismic table performance at LNEC comes in the form of pseudo-open loop control, whereby, before every test run, a proprietary software called *Adapt*, through a set of user-defined parameters and decisions, identifies the system transfer function, and then modifies the reference signal, creating what we will now refer to as *driver signal*, which is then used as the reference in the closed-loop control of the seismic table position.

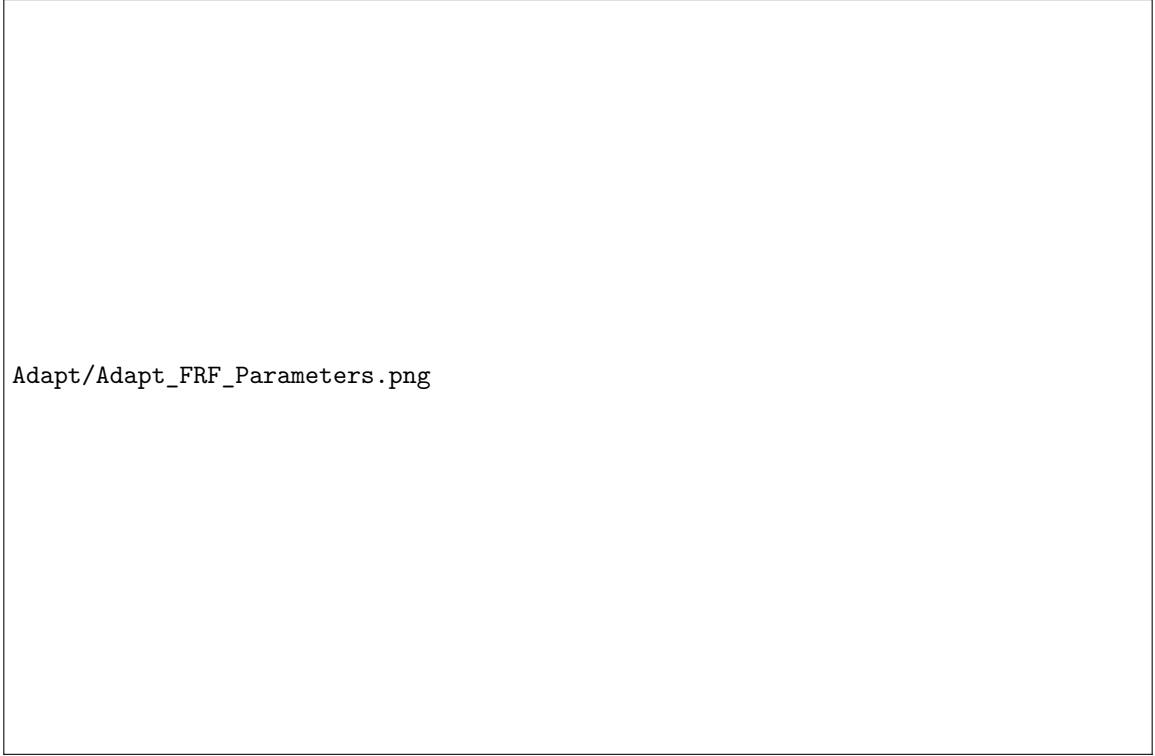
In the system identification menu (figure 74) the user can apply various filters to the acquired displacement and acceleration signals, and then choose how to synthesize both into a displacement signal. This is done to reject the inaccurate readings of position sensors at high frequencies, and of acceleration sensors at low frequencies, and combine the information of both to generate a more accurate displacement signal. Then the user can choose various options for the computation of the transfer function.

Adapt/driver_signal_generation_loop.png

Figure 72: A simplified block diagram of driver signal modification loop

Adapt/Adapt_Project_Parameters.png

Figure 73: Adapt Project Parameters menu - In this menu the user configures which files were used for determining the system's FRF, and the reference (or target) signal.



Adapt/Adapt_FRF_Parameters.png

Figure 74: Adapt FRF Parameters menu - The modification of the driver signal requires the identification of the frequency response function of the system. In this menu the user sets parameters which are used in the composition of a displacement signal from acceleration and displacement data acquired from a test run where 3 minutes of pink noise are used as input. The user also defines the settings for the computation of the FRF.



Adapt/Adapt_INV_Panel.png

Figure 75: Adapt INV Panel menu - This menu is used to generate new driver signals, and takes as inputs the original reference (referred to as *target*), the driver signal used in the test run, it's measured output at the seismic table, and user defined parameters which affect which parts of the signal are modified, and by how much they're modified.

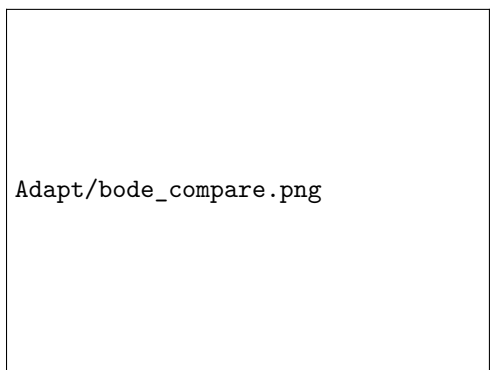
4.2.3.2 Simulating the driver adaptation process

For this section the coupled 2-dof system had both it's masses set to 2000 kg, $f_1 = 1.5Hz$, $\zeta_1 = 0.02$, $f_2 = 6Hz$, $\zeta_2 = 0.05$, and the optimal controller with $Q = 10^3$ (as in Figure 52). The code implementation can be found at the following GitHub repository location.

4.2.3.3 System Identification

The first step in the driver adaptation process is identifying the system transfer function (seismic platform + model). To this end, the seismic table is run for 3 minutes with a pink noise signal as input.

This was simulated by converting a pink noise *.drv* file (a proprietary file format meant for use with LNEC proprietary software tools: *LNEC SPA* and *Adapt*) used in one of LNEC's seismic tests, through the use of a Python library which replicates the functioning of LNEC SPA, which allows the conversion to standard text file, and then providing this signal as input to the MATLAB model previously developed. The simulation output, seismic table position and acceleration over time, is then written to a *.acq* file, again using *LNEC SPA* Python library functions. The driver and simulated acquired data files are then input to *Adapt*, where the system frequency response function is determined.



Adapt/bode_compare.png

Figure 76: Identified frequency response function from *Adapt* (using "bad" settings) versus the model from Section 4.1

4.2.3.4 Generating the target files (.tgt)

The acceleration signal of each seism was integrated twice using a simple trapezoidal rule. From the obtained displacement signal, the initial and final displacement values were used to compute the average velocity, which was then used to subtract the drift from displacement signal, and have it begin and end at zero. Then the signal was scaled to fit the displacement range of seismic table (and the mathematical model).

4.2.3.5 Generating the adapted driver

The first driver (driver zero) was generated using just the target file. This driver was then converted to text format using the previously mentioned Python functions, and then input into the Matlab model. From the displacement output, the acceleration was computed using a second order central difference. This data was then written to .acq format, and used to generate driver 1 in *Adapt.exe*. The process was repeated to generate subsequent drivers.

4.2.3.6 Performance Analysis

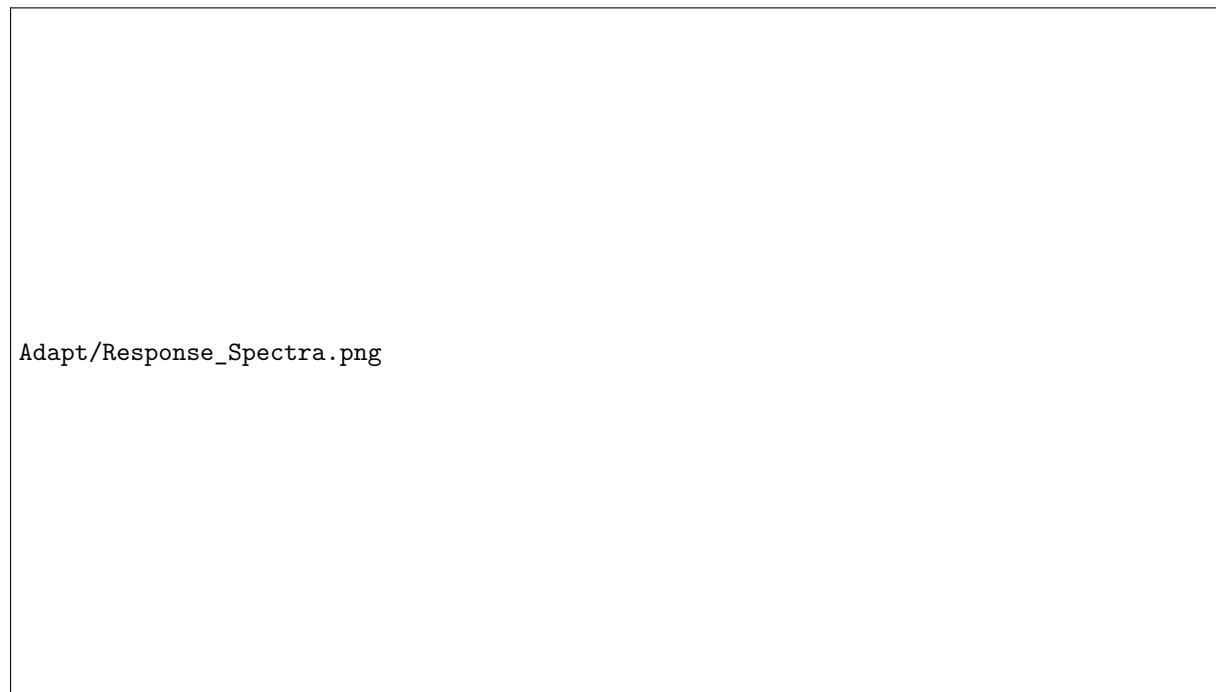


Figure 77: Target response spectrum and the resulting response spectrum from tuned PIDF, optimal controller, and the 3 iterations of adapted drivers resulting from FRF shown in figure 76.

For the seismic signal tested in this section (L'Aquila reduced scale seism), the response spectra from adapted drivers underperforms compared to the PIDF controller, and even more so when compared to the optimal controller.

In figure 80, the identified FRF (figure 78) used for adapting the drivers is closer to the model, resulting in a better adaptation of the driver signal, but it still requires 3 iterations to somewhat achieve the target response spectrum, and still doesn't perform as well as the optimal and PIDF controllers.

In figure 77, the identified FRF (figure 76) used for adapting the drivers isn't as good as in the other case (figure 78), due to the settings defined by the user in *Adapt*, and the resulting response spectrum after three iterations isn't very satisfactory. This shows the one of limitations of this method: It depends on the user to make the right decisions, but the user isn't perfect and there is no guarantee he'll opt for the best settings. Another limitation is the fact that the user does not have exact knowledge of the best settings, because unlike in this example, where the system is known and the FRF is made to match

it (Figure 78), in reality this is not possible, leaving the user to make an informed guess based on the information displayed in *Adapt* and past experience.

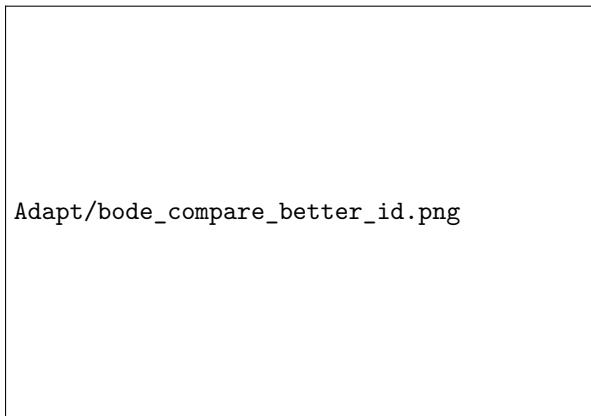


Figure 78: Frequency response function of identified system using 1024 Low Sidelobe windows results in a rougher approximation of the transfer function, but handles noisy data better.



Figure 79: Frequency response function of identified system using 20 Hanning windows results in a much better approximation of the transfer function, but likely wouldn't work as well if using noisy sensor data.



Figure 80: Target response spectrum and the resulting response spectrum from tuned PIDF, optimal controller, and the 3 iterations of adapted drivers resulting from FRF shown in figure 78.

Adapt/excel_accel_resp_spectrum.png

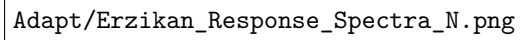
Figure 81: Acceleration response spectrum MSE for various seismic signals. (For the Erzikan sism (figures 83 & 84), an additional driver adaption iteration was performed, and the results shown are for the resulting 4th driver)

Adapt/excel_disp_resp_spectrum.png

Figure 82: Displacement response spectrum MSE for various seismic signals. (For the Erzikan sism (figures 83 & 84), an additional driver adaption iteration was performed, and the results shown are for the resulting 4th driver)

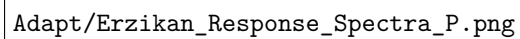
Control Strategy	PIDF	Optimal Control	Adapted Driver (3rd)
Mean Acceleration MSE (m/s^2)	8.27×10^{-2}	2.09×10^{-2}	4.17×10^{-1}
Median Acceleration MSE (m/s^2)	3.20×10^{-2}	4.39×10^{-4}	9.48×10^{-2}
Mean Displacement MSE (m)	5.97×10^{-7}	3.01×10^{-11}	1.62×10^{-5}
Median Displacement MSE (m)	2.51×10^{-7}	9.42×10^{-12}	1.73×10^{-5}

Table 4: Comparison of mean and median response spectrum MSE for various seismic signals.

The plot area is empty, showing only the axes and the filename.

Adapt/Erzikan_Response_Spectra_N.png

Figure 83: Erzikan fault normal response spectra. For this case a 4th adapted driver was computed, which yielded better results than the PIDF (unlike the 3rd driver used in every other performance comparison), but only in terms of acceleration response spectrum and in the fault normal direction.

The plot area is empty, showing only the axes and the filename.

Adapt/Erzikan_Response_Spectra_P.png

Figure 84: Erzikan fault parallel response spectra. In this case the adapted drivers perform very poorly for frequencies above 10Hz.

4.3 Conclusions: PIDF vs Optimal Control vs Driver Iteration

Regarding the analysis in sections 4.2.1 and 4.2.2, optimal control results in a much more precise replication of the ground response spectra at low frequencies, unlike the tuned PIDF which exceeds the desired response spectra in the same band. However, the trade-off for this accuracy at lower frequencies is a response spectrum that falls short at higher ones. The optimal controller can be tuned for increasingly larger cut off frequencies without undesired amplification at low frequency, unlike the PIDF, but the results of this in practice may differ from the simulation studies performed using the current model.

The results indicate that the tuned controllers (PIDF & Optimal) offer superior performance in terms of response spectra compared to the adapted drivers, while also requiring less user intervention doing so, with optimal control providing the best results in every instance. However, both the performance of the controllers and the adapted drivers depends on identifying the system, and the sensitivity of each method to quality of the identification is still unknown.

5 Experimental Validation

In order to experimentally validate the performance of the control strategies studied in 4, the two controllers were implemented in the LabVIEW software running in real time on

5.1 Measurement and Control Chain

Using the LNEC mini shaking table. This equipment consists of an electromagnetic shaker to input the motion to the platform, which rolls over the fixed body. The accelerometers for characterization are attached to the platform. This equipment has a displacement transducer (LVDT type) embedded used to measure the platform displacement



Figure 85: measurement and control chain schematic

5.1.1 Calibration of Displacement Sensor

ver excel no computador da mesa sismica

5.1.2 Calibration of Acceleration Sensor

1. In the acquisition/control software, set scaling to 1 and offset to zero
2. Place accelerometer vertically on a flat stationary surface (aligning the accelerometer measurement axis orthogonally to the acceleration of gravity) such that it will measure zero acceleration
3. Set the offset in the signal conditioner such that the output is zero

4. Place the accelerometer horizontally on a flat stationary surface, aligning accelerometer measurement axis with the acceleration of gravity, such that it will be sensing 1G
5. Set scaling in the conditioner and the software such that the output is 1G

5.2 Identification of Electrodynamic Shaker

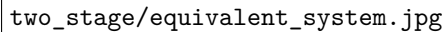
In order to identify a model of the shaker+mini table system, white noise signals of different amplitudes were used as input. Two controller settings were also used, one using just proportional control with $P=1$, and another with proportional and integral control, with $P=10$ and $I=1$.

- **Simplicity:** Applies typical open-loop identification methods through a reformulation of the problem.
- **Flexibility:** The controller may be nonlinear and unknown.
- **Consistency:** Good identification of the plant is independent of the noise modelling.
- **Robustness:** Consistent identification of S_0 is not a requirement for good plant identification; a high variance of the S_0 model is acceptable.

In order to tune the controller to the system parameters, it is necessary to identify the controlled system. For this purpose, an identification experiment was carried out using pink noise with frequency content up to 40 Hz, and a controller with proportional gain. To address the particularities of closed-loop identification, the “two-stage method” was adopted.

5.2.1 Applying Two-Stage Identification Method

The code implementation can be found at the following GitHub repository location.



two_stage/equivalent_system.jpg

Figure 86: Cap

1. Identification of the sensitivity function S_0 :

- A linear FIR model with 100 parameters was fitted (the number of parameters was chosen arbitrarily for this stage).
- A Hammerstein-Wiener model was then defined, using the previously identified FIR model as the linear block, and with the output nonlinearity equal to that of the physical system (saturation of the control action at known values), resulting in a refined model.

2. Reconstruction of the control action isolated from noise effects ($\hat{u}_r$).

3. Identification of G_0 :

- Non-parametric identification indicated a 4th-order model.
- A model with 4 poles and 3 zeros was fitted using the reconstructed control action and the measured output.

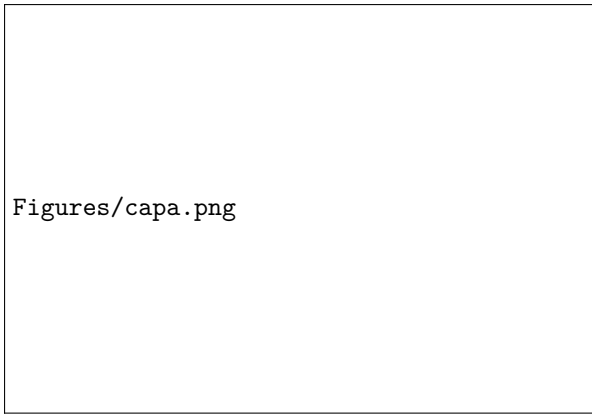


Figure 87: Cap

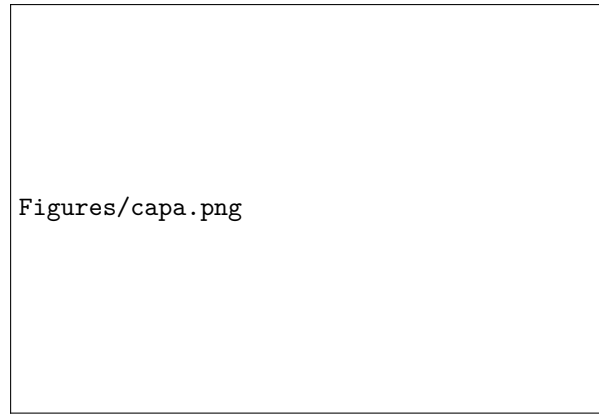


Figure 88: F

5.2.2 Checking the shaker and table for mechanical issues

Given the unexpected nonlinearities found during the identification (figures 87 and 88), the power to the shaker was turned off and the table was directly actuated by hand through its entire travel, and with direction changes, in order to feel for any binding or play in the system. Since binding could be felt at certain points in the travel, both the shaker and the table slideways were disassembled to check for issues.



Figure 89: Caption

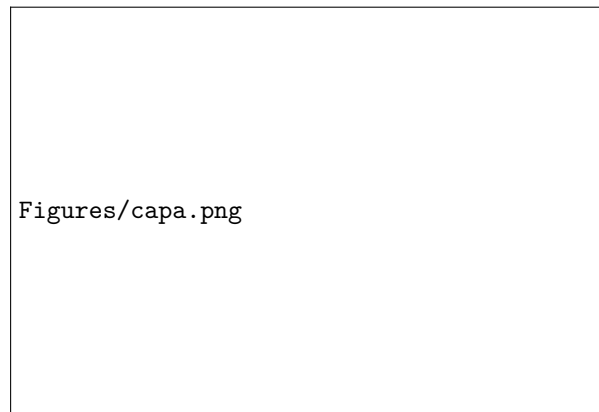


Figure 90: another Caption 2

With the table disconnected from the shaker, it was once again actuated by hand and appeared to be running smooth, without any binding or play. Regardless, the slideways were cleaned, visually inspected, lubed, and reassembled.

With the table disconnected from the shaker, while actuating the shaker shaft by hand through its travel, often times binding could be felt. With the shaker disassembled, it was verified that a flat helical spring which connects the actuator shaft assembly to the shaker chassis had the come loose, and this may have been the source of the issue. Given the interface materials, cyanoacrylate glue was deemed suitable for the job, and the spring was bonded in its original place. No other obvious issues were found.



Figure 91: imagem do interior do shaker



Figure 92: imagem das guias da mesa

Everything was reassembled together, and new data was collected for another system identification, however, a nonlinearity was still found.

5.3 Implementar PIDF e controlo ótimo na aplicação de LabView de controlo das mesas

$$C(z) = K_p + K_i \frac{T_s}{z-1} + K_d \frac{1}{T_f + \frac{T_s}{z-1}} \quad (5.3.1)$$

5.4 Experimental Results

Once the system was identified, the PIDF parameter tuning algorithm was run, defining three different bandwidths for the closed-loop response: 10Hz, 15Hz, and 20Hz. The three sets of parameters were then tested against the iterative drive signal correction process. For this seismic signal, it was found that all tuned controllers performed better than the 5th driver of the iterative process.

For this other seismic signal, it was found that all tuned controllers performed better than the third driver in the iterative process. However, the fifth driver already performed better. In these two tests, acceleration data were not measured, so the displacement data were extrapolated to determine the acceleration response spectrum.

5.5 Further Simulation Studies (Robustness)

- ⇒ Avaliar robustez do método de indentificação e sintonização do controlador
- ⇒ Simular dados com o modelo da mesa uniaxial, desta vez introduzindo uma perturbação na saída

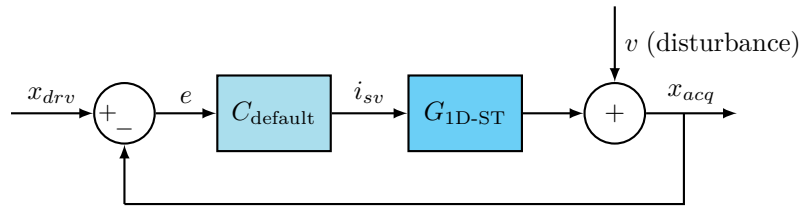


Figure 93: Closed-loop System

- ⇒ Fazer identificação $\hat{G}(z)$ a partir dos dados simulados
- ⇒ Sintonizar controladores para $\hat{G}(z)$
- ⇒ Verificar desempenho dos controladores, fazendo novamente simulação para o modelo 1DST com perturbações na saída

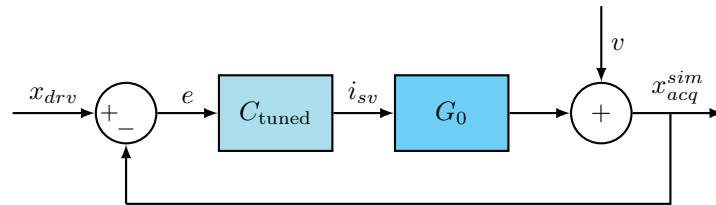


Figure 94: Closed-loop System

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